

Tunable Light Source Setup

User Manual

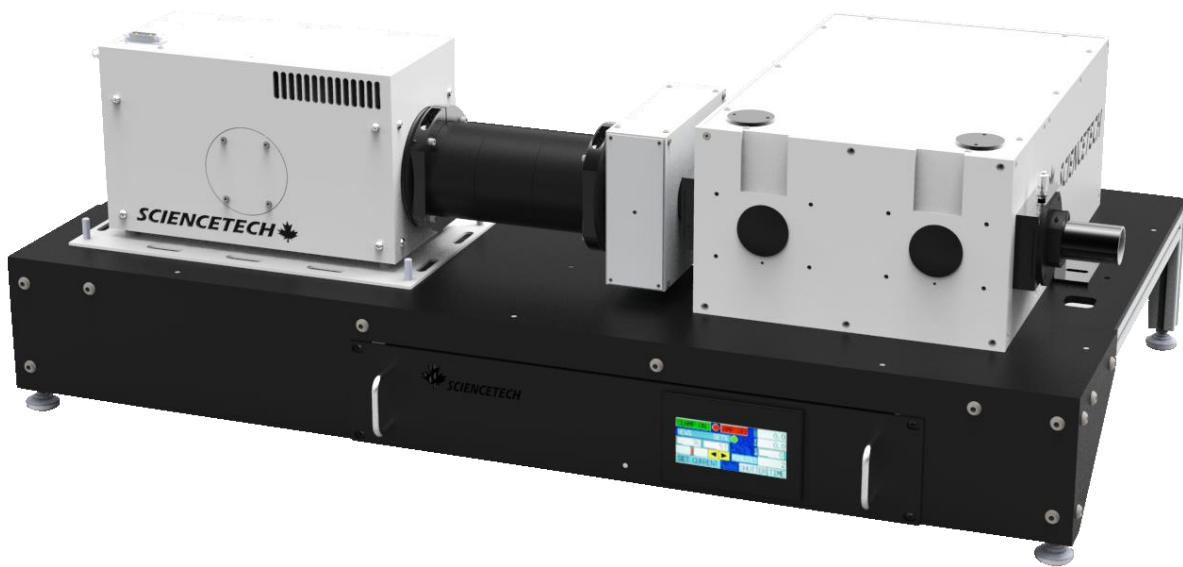


Table of Contents

1. TLS System Overview	7
1.1 TLS System Overview	7
1.2 TLS System Configurations	7
2. Safety Instructions	10
2.1 List of Specific Hazards	10
2.1.1 High Intensity UV and Visible Radiation	10
2.1.2 Lamp Explosion.....	11
2.1.3 Ozone	11
2.1.4 Electrical Shock.....	12
2.1.5 Electromagnetic Interface (EMI)	12
2.1.6 Heat	13
2.1.7 Interlocking System	13
2.2 General Warning Symbols.....	13
2.2.1 General Warning Caution	13
2.2.2 Electrical Shock.....	14
2.2.3 Hot Surface	14
2.2.4 Alternating Voltage Symbol	14
2.2.5 Waste Electrical and Electronic Equipment	14
2.2.6 ON.....	14
2.2.7 OFF.....	15
2.3 Suggested Personal Protective Equipment.....	15
2.3.1 PPE When Handling the Lamp	15
2.3.2 PPE for Normal System Operation	15
2.3.3 PPE for Normal System Operation	15
2.3.4 List of Suggested PPE Equipment.....	16
3. Installation Overview	19
4. Shipping and Unpacking.....	20
4.1 Unpacking	20
4.2 Handling	21
4.2.1 Non-DUAL System	21
4.2.2 Handling TLS-55-DUAL.....	22
4.3 Included Components.....	23
4.3.1 Installation of F/3 Collimating Optic Lens Tube.....	26

5.	Instructions for Arc Lamp Systems	27
5.1	General Xenon Arc Lamp Information	27
5.2	Mechanical Handling	28
5.3	Xenon Arc Lamp Installation.....	30
6.	Instructions for QTH Lamp Systems	32
6.1	General QTH Lamp Information	32
6.2	QTH Lamp Installation.....	33
7.	Instructions for IR Source Systems.....	34
7.1	General IR Source Lamp Information	34
7.2	IR Source Installation	35
8.	System Connections.....	37
8.1	Interconnection Table	37
8.2	System Connection Instructions	38
8.2.1	Standard TLS	38
8.2.2	TLS-55-DUAL.....	39
9.	General Dimensions	40
10.	Power Supply Description and Operation.....	45
10.1	601 Series Power Supply Overview and Connections	45
10.1.1	Back Panel Connections	45
10.1.2	Touchscreen Types.....	46
10.2	Control Unit Operation for Arc Lamp Systems (601- Power Supply Type I)	47
10.2.1	Power Supply Start-up.....	47
10.2.2	Main Screen	48
10.2.3	Starting the Lamp	49
10.2.4	Log Screen.....	50
10.2.5	Shutter Screen	51
10.3	Control Unit Operation for Arc Lamp Systems (601- Power Supply Type M)	52
10.3.1	Power Supply Start-up.....	52
10.3.2	Main Screen	53
10.3.3	Starting the Lamp	54
10.3.4	Menu Screen.....	54
10.3.5	Statistics Screen.....	55
10.3.6	Shutter Screen	56
10.4	Control Unit Operation for QTH and IR Source Systems (602 Power Supply)	57
10.4.1	Main Screen	58

10.4.2	Setup Screen	59
10.4.3	Starting the Lamp	60
10.4.4	Time Screen.....	60
10.4.5	Shutter Screen	61
11.	Software Installation.....	62
12.	General System Operation.....	63
12.1	Setting System Bandpass	63
13.	Overall System Specifications.....	65
14.	EU Declaration of Conformity.....	67
15.	Troubleshooting	68
16.	Accessories	69
16.1	Off-axis Elliptical Reflector Installation (OAE)	69
16.2	Dual Input (DPIN-ST-MAN) and Output (DPOUT-ST-MAN) Ports for Sciencetech Standard Monochromators	69
17.	Appendix B. Power Optimization.....	72
17.1	Power Optimization in TLS-72-X300 and TLS-55-X300 and 37W IR Source	72
17.2	Power Optimization in TLS-72-Q250 and TLS-55-Q250	74
18.	Appendix C. Operating Adjustable Iris to Modify the Size of the Output Beam,	75
19.	Important Notice.....	78
20.	Warranty and Assistance	80

Table of Figures

Figure 1.	Standard TLS-55-300X (top) and (TLS-72-Q250) System.	8
Figure 2:	Schematic view of example TLS-55-DUAL with accessories. Your system may not include all accessories depending on your system configuration.	9
Figure 3.	Radiation hazard category of the indirect beam from rear of the -Q250 and -37IR TLS (Left) and for -X300 TLS (right).	10
Figure 4.	Radiation hazard of the direct white light beam (total specular reflectance from the grating surface) of -X300 (right) and -Q250 and -37WIR are categorized as RISK GROUP 3.	11
Figure 5.	Disposable Nitrile Glove.	16
Figure 6.	Outdoor Skin Protection.....	16
Figure 7.	Face Shield with Protective Coating.	17
Figure 8.	Sunglasses.....	17
Figure 9.	Comfort-Fit Wraparound Safety Glasses.	18
Figure 10.	Lifting locations of TLS system.	21
Figure 11:	Lift points for a TLS-55-DUAL.	22
Figure 12.	Different TLS systems overview.....	23
Figure 13.	Power supply cable with IEC 320-C13 termination.....	23

Figure 14. Xenon Short Arc Lamp.	25
Figure 15. 250W QTH Lamp.....	25
Figure 16: 37W IR Filament.....	25
Figure 17: Attaching Collimating Optic 3" Lens Tube to Output Slit of Monochromator.	26
Figure 18. A Xenon Arc lamp.....	27
Figure 19. Blue M4x0.7 SHCS screws at the rear panel of TLS-55-X300 and TLS-72-X300.	30
Figure 20. Installed xenon arc lamp into a LH-E lamp stage.	31
Figure 21. A Standard QTH Lamp.	32
Figure 22. Removing the lamp stage form the LH-S lamp housing of TLS-55-Q250 and TLS-72-Q250.....	33
Figure 23. Inserting QTH lamp into the LH-S lamp housing.	33
Figure 24. An IR Source.	34
Figure 25. Blue M4x0.7 SHCS screws at the rear panel of the lamp house.	35
Figure 26. Installation of IR Source.....	36
Figure 27: Connection location figure.	38
Figure 28: Power entry and USB on the rear of a TLS-55-DUAL. 7W2 cable connection example. Monochromator and filter wheel connections.....	39
Figure 29. Dimensions of TLS-55-X300.....	40
Figure 30. Dimensions of TLS-72-X300.....	41
Figure 31. Dimensions of TLS-55-Q250.	42
Figure 32. Dimensions of TLS-72-Q250.	43
Figure 33: TLS-55-DUAL dimensions.	44
Figure 34. Rear of 601 series power supply with connections labelled.	45
Figure 35. 601 Front Panels showing Type I (top) and Type M (bottom) touchscreens.....	46
Figure 36: Opening screen (Type I PLC)	47
Figure 37: Main menu (Type I PLC).....	48
Figure 38: Log screen (Type I PLC).....	50
Figure 39: Shutter screen (Type I PLC)	51
Figure 40: Opening screen (Type M PLC)	52
Figure 41: Main screen (Type M PLC)	53
Figure 42: Menu screen (Type M PLC).....	54
Figure 43: Statistics screen (Type M PLC).....	55
Figure 44: Shutter screen (Type M PLC)	56
Figure 45: Start screen (602 series power supply).....	57
Figure 46: Main screen (602 series power supply).....	58
Figure 47: Setup screen (602 series power supply)	59
Figure 48: Time screen (602 series power supply)	60
Figure 49: Shutter screen of 602 series power supply	61
Figure 50. Slit with micrometer adjust.	63
Figure 51. EU declaration of conformity for TLS systems.	67
Figure 52: OAE installation.	69
Figure 53. Push down the knurled knob the knob all the way down.	70
Figure 54. Turn the knob until the arrow on the knob align with the FRONT arrow on the label. You will feel a resistance at this location.	70
Figure 55. Slowly release the knob.....	70
Figure 56. The rear of the TLS showing the three M3 lamp alignment screws.....	73
Figure 57. The rear of the TLS with QTH lamp showing the three M3 lamp alignment screws of the spherical back reflector.....	74

Figure 58. Technical drawing of the THORLABS SM1D1D iris indicating the location of the locking screw.	75
Figure 59. Technical drawing of the THORLABS SM1D1D iris indicating the location of the locking screw and the graduated ring with laser-engraved numbers.	76
Figure 60. Unlocking and changing the aperture size of the adjustable iris to modify the size of the focused beam.	77

Table of Tables

Table 1. Sciencetech TLS systems breakdown. For a DUAL system, 2 lamp types must be chosen at time of ordering.	7
Table 2. Interconnection table.....	37
Table 3. Label acronyms, descriptions and connector types explained.....	45
Table 4: Micrometer slit width adjustment table.	63
Table 5. Specifications and Certificates of Tunable Light Source systems.....	65

1. TLS System Overview

1.1 TLS System Overview

Sciencetech Tunable Light Sources (TLS) provide narrow bandwidth light at wavelengths within their operating range by selecting the desired component from the output of a broadband light source (Xenon, QTH, or IR Source) using a monochromator. The TLS also contains a filter wheel between the broadband light source and the monochromator to remove unwanted harmonics.

1.2 TLS System Configurations

Sciencetech TLS systems are modular instruments. They are constructed using Sciencetech LH series lamp houses. The 300W Xenon Arc lamps and 37W IR Filament are installed in LH-E lamp housings with elliptical reflectors. The 250Q Quartz Tungsten Halogen lamps are installed in LH-S lamp housings with back spherical reflectors. The -72 systems include a 1/8-meter focal length Czerney-Turner monochromator. The -55 systems include 9055 ¼ meter focal length Czerney-Turner monochromator. -55 monochromators can achieve twice as high optical throughput than 9072 monochromators. Each monochromator includes adjustable slits at the entrance and exit of the system which provides control over the optical bandpass. Each TLS system includes a filter wheel loaded with hard coated thin film interference long pass filters to remove undesired harmonic wavelengths from the optical output. The operation of the components is powered and controlled from a control unit. Xenon arc lamp systems include 601 series control units. QTH and IR Filament lamp systems include 602 series control units. All TLS systems are fully customizable. Contact Sciencetech if you would like to modify or customize your system.

Table 1. Sciencetech TLS systems breakdown. For a DUAL system, 2 lamp types must be chosen at time of ordering.

Product Line	Model	Monochromator	Lamp Type
TLS	TLS-72-300X	9072 (1/8 m)	-300X
TLS	TLS-72-250Q	99072 (1/8 m)	-250Q
TLS	TLS-55-300X	9055 (1/4m)	-300X
TLS	TLS-55-250Q	9055 (1/4m)	-250Q
TLS	TLS-55-DUAL	9055 (1/4 m)	-300X -250Q -37IR

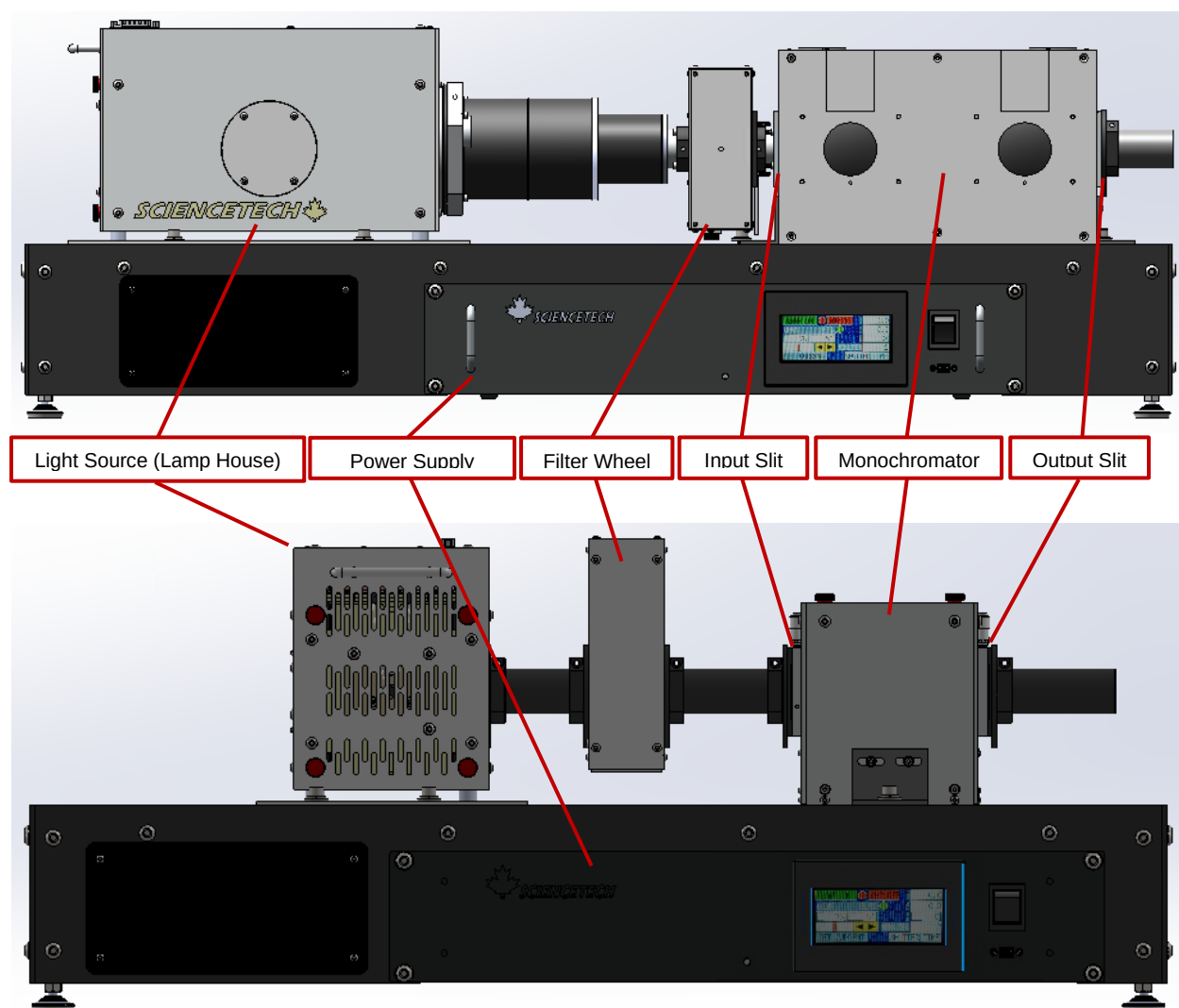


Figure 1. Standard TLS-55-300X (top) and (TLS-72-Q250) System.

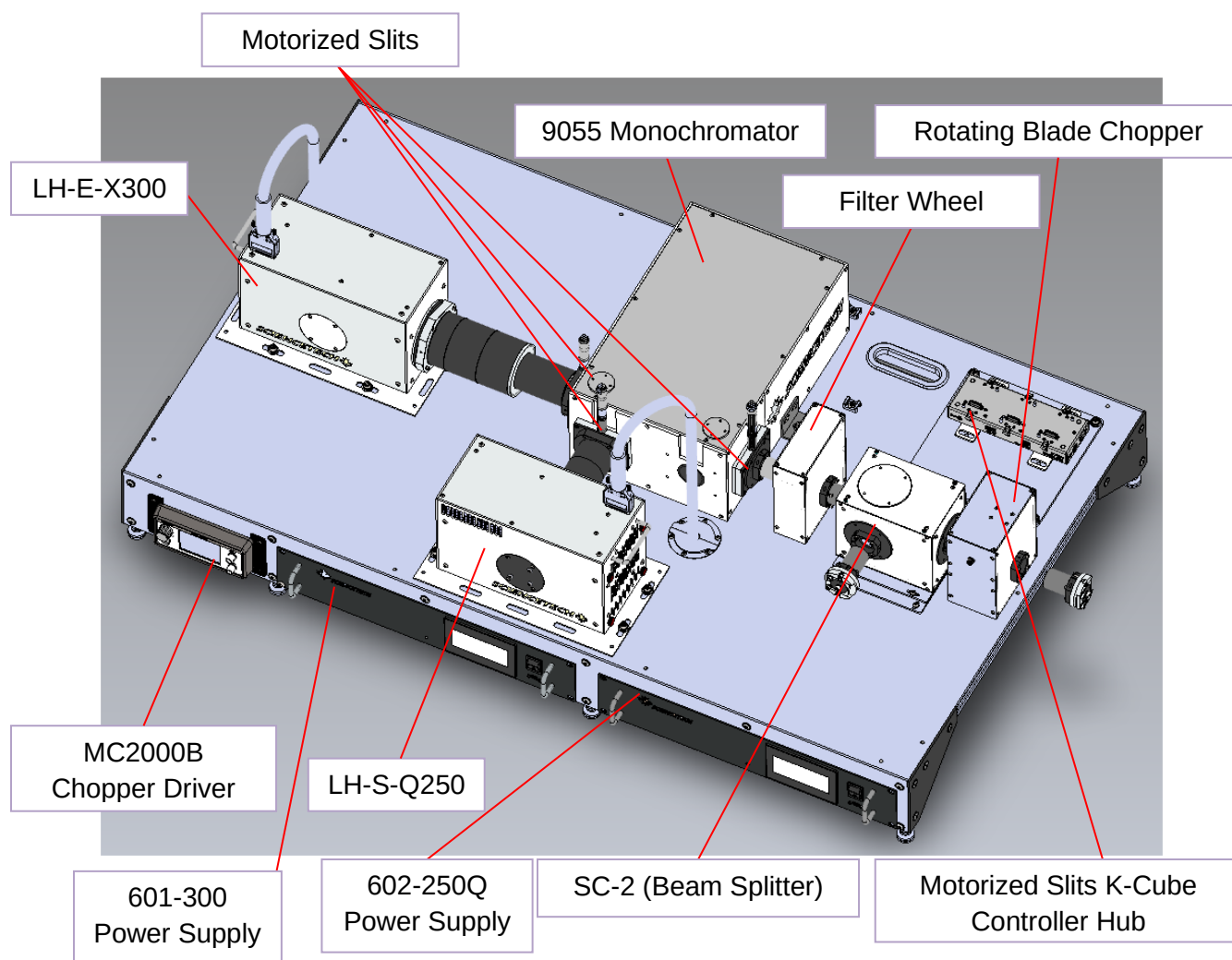


Figure 2: Schematic view of example TLS-55-DUAL with accessories. Your system may not include all accessories depending on your system configuration.

2. Safety Instructions

2.1 List of Specific Hazards



The following is a list of the most important hazard associated with using TLS series light sources. Make sure to review these hazards and take appropriate steps to avoid them. This information **MUST** be reviewed in all cases where the symbol shown above is marked on the equipment, in order to find out the nature of the potential hazards and any actions which have to be taken to avoid them.

2.1.1 High Intensity UV and Visible Radiation



TLS series light sources can output high intensity radiation in UV, visible, and IR regions of the electromagnetic radiations. The high intensity UV, visible, IR radiation can cause severe eye injury.

Safety Precautions:

- Never look directly at an operating lamp.
- Always wear UV protective goggles such as welder's goggles or appropriate UV sun glasses when working with the operating xenon arc lamps or QTH lamps.
- Do not place hands or combustible materials in the path of the light beam.

According to IEC/EN 62471 the indirect light (stray light) emitted from back of the Q250, X300, and 37IR is RISK GROUP 2. The hazard category and recommended preventive measures for the stray light of each system are shown in Figure 3. The safe distance away from the rear of the -Q250 and IR37 are 30 cm and for -X300 is 70 cm.

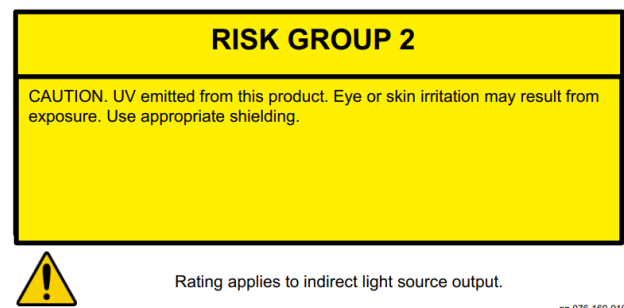
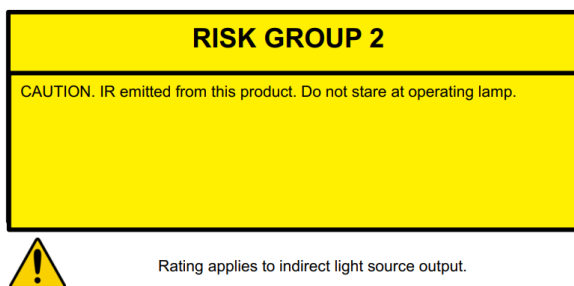


Figure 3. Radiation hazard category of the indirect beam from rear of the -Q250 and -37IR TLS (Left) and for -X300 TLS (right).

The direct beam of the -X300, -Q250, and -IR37 at total specular reflectance (white light) from the grating (shown as 0 nm in SciSpec software) is categorized as RISK GROUP 3.

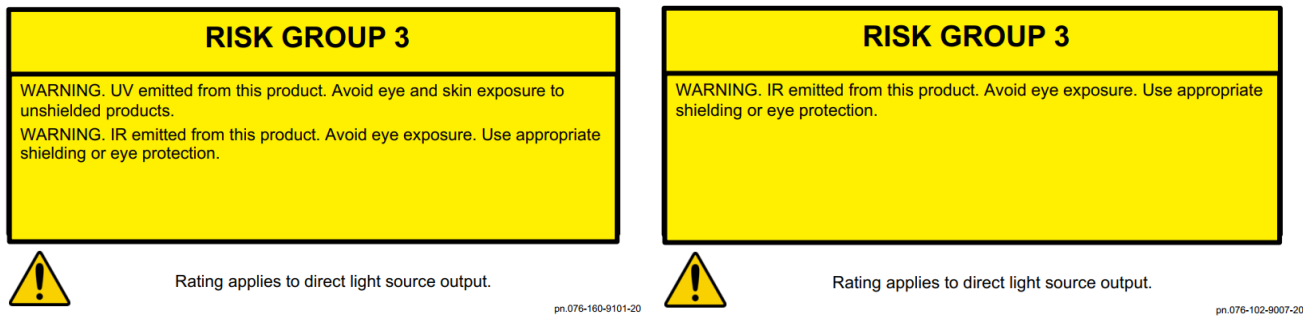


Figure 4. Radiation hazard of the direct white light beam (total specular reflectance from the grating surface) of -X300 (right) and -Q250 and -37WIR are categorized as RISK GROUP 3.

2.1.2 Lamp Explosion

A short arc lamp (including Xenon short arc lamps used in TLS-55-X300 and TLS-72-X300 product) contains high pressure gases up to 30 bar (even more in low-wattage lamps). In case of damage to the lamp envelope, there is enough energy stored in the lamp bulb to send quartz splinters flying several meters across a room. While their envelope is made of robust quartz or other types of glasses, they need to be handled carefully. They must be protected against shock, impact and excessive force. QTH lamps used in TLS-55-Q250 and TLS-72-Q250 products are not highly pressurized compare to xenon short arc lamp and thus expose less risk of explosion.

Certain precautions from the lamp manufacturer and Sciencetech must be observed when handling them.

Safety Precautions:

- Always wear safety goggles (face shield preferred) while handling the lamp.
- Protect other exposed areas of your body such as neck and hands with appropriate protective equipment.
- Always wear gloves when handling the lamp. Fingerprints or other contamination on the lamp envelope can create a hot spot during the lamp operation which could cause an explosion or damage to the lamp.
- Never touch the lamp envelope. Always handle the lamp by holding its cathode and anode.
- Before installing the lamp, always store the lamp in its original box. Never place the lamp on hard surfaces, such as a desktop, as this can damage the lamp envelope.
- Arc lamps are often supplied in a safety cover. This protects users from possible spontaneous or induced bursting of the lamp during handling. Thus, if applicable, remove the safety cover only after installing the lamp at the cathode and anode holder. Make sure to remove the cover before turning on the lamp.
- The majority of the arc lamps contain a thin external wire along the external surface of the lamp envelope. This wire is necessary for initial ignition of the lamp. Never tamper with this wire.

2.1.3 Ozone

Unless otherwise stated by Sciencetech, the xenon arc lamp that you have received from Sciencetech is an ozone-free lamp. The QTH lamp provided with TLS-55-Q250 and TLS-72-Q250 poses no risk with regard to ozone emission.

Safety Precautions:

- Always use ozone-free xenon arc lamp. It is always advised to contact Sciencetech before purchasing a spare lamp.
- Inhaling ozone can cause permanent damage to the lungs. Lower concentration of ozone can cause shortness of breath, headache, chest pain, lung and throat irritation and other temporary or permanent damages to the body. If you are operating an ozone producing arc lamp, use connect an ozone convertor to create a safe operating environment.
- If you are provided with an ozone producing arc lamp, make sure to place the system under well ventilated fume hood which prohibit release of any ozone gas to your working area.
- Use an ozone meter to monitor the ozone level of the room. Please contact Sciencetech for recommendation regarding the appropriate ozone meter.

2.1.4 Electrical Shock



TLS power supplies are powered by 100-230 VAC. In addition, during the initial ignition of the lamp in TLS-55-X300 and TLS-72-X300, up to 25 KV DC is applied to the xenon lamp for few microseconds before maintaining constant current to the lamp. Thus, there is a great danger of electric shock while working with the system and extreme care is required while working with the system.

Safety Precautions:

- Disconnect the AC supply before servicing a TLS or its power supply.
- Disconnect the AC supply before inserting or removing the lamp.
- Avoid touching the exposed electrical terminal of the system.

2.1.5 Electromagnetic Interface (EMI)



Operation of an arc lamp (in TLS-55-X300 and TLS-72-X300) requires a high ignition voltage to create a plasma in the lamp envelope and a high current to sustain this plasma between the anode and cathode. These events can create EMI. While the TLS series light source lamp housing and power supply are designed to contain these EMI, additional care is required to protect personnel and other surrounding electronic equipment.

Safety Precautions:

- Keep all electronic equipment, such as computers, more than 2 feet away from the TLS.
- Start the arc lamp before turning on other electronic at the vicinity of the TLS.

- Connect the TLS's power supply to a separate outlet and line from other equipment in your lab.
- Only connect the power supply to electrical outlets with a ground.
- If you are using a pacemaker or ICD consult with your physician regarding the danger the possible danger that working with arc lamp power supply might have for you.

2.1.6 Heat



Lamps operate at very high temperatures. Thus, efficient cooling is required during their operation and several minutes after they have turn off.

Safety Precautions:

- Always make sure the fans on the lamp housing and the power supply are running during the lamp operation.
- Let the fans operate for five minutes after turning the lamp off.
- Always assume the lamp is hot.
- If you are removing the lamp stage from the lamp housing, monitor the heat stickers that are placed on the anode and cathode of the lamp (if applicable). If any of the dots on these stickers turn black, this means the lamp is overheating. Contact Sciencetech before using the system again.
- Always make sure there is at least 5-inch clearance near the fans on the power supply and the lamp housing. Do not stack items on top of the power supply.
- Operate the TLS only indoor, in a clean, cool, and well-ventilated room.

2.1.7 Interlocking System

The interlocking system is a secondary step to protect the system.

Safety Precautions:

- Never defeat the interlocking system or tamper with it.
- The customer is not supposed to perform service on the system. Always contact Sciencetech for any questions regarding the performance of the system.

2.2 General Warning Symbols

The following symbols are used both on this documentation and also posted on the TLS series light source.

2.2.1 General Warning Caution



The exclamation symbol appears above, indicates warning or caution for matter that might cause harm or damage to the personnel or equipment. Refer to user manual for more instructions.

2.2.2 Electrical Shock



The electric shock symbol appears above, indicates hazards to personnel or equipment that may cause due to the high voltage.

2.2.3 Hot Surface



Hot surface symbol appears above, indicates presence of any hot surface, internal hot competent, or required cooling.

2.2.4 Alternating Voltage Symbol



The above sign is the international symbol of alternating voltage (VAC) or current.

2.2.5 Waste Electrical and Electronic Equipment



The above symbol on this product or its packaging indicates that this product should not be disposed with regular waste. It is the responsibility of each user to dispose this item based on their local regulatory law. Sciencetech is committed to provide consultation about disposal and recycling of its products.

2.2.6 ON



The ON sign indicated the ON position of the power switch.

2.2.7 OFF



The OFF sign indicates the position of the OFF button on the power switch.

2.3 Suggested Personal Protective Equipment

2.3.1 PPE When Handling the Lamp

The lamp should be turned off and power to the power supply should be turned off and fully disconnected from the wall plug.

If you will be handling or observing the lamp, such as lamp installation, it is recommended that the following PPE be used.

Eyes	Face shield with protective coating or safety goggles
Face	Face shield with protective coating
Hands	Powder free Nitrile gloves
Arms	Clothes that leave no gaps between gloves and shoulders
Neck	Covered by face shield and clothing

2.3.2 PPE for Normal System Operation

Any system alignment with the lamp on is hazardous. The user should NEVER remove any access covers with the lamp on. Contact Sciencetech if you suspect that the system is out of alignment, or to understand what is required to measure system alignment.

2.3.3 PPE for Normal System Operation

It is recommended that sunglasses or protective safety glasses (with UV protective lenses) be worn to protect against stray radiation. There is stray radiation from the grill at the rear of the system (Section 2.1.1). Never look directly into the rear of the system for an extended period of time.

Eyes	Sunglasses
Face	Recommended
Hands	Recommended
Arms	Recommended
Neck	Recommended

2.3.4 List of Suggested PPE Equipment

The following PPE items are all readily available from mcmaster-carr.com and are shown with their mcmaster-carr.com product numbers.

Disposable Nitrile Glove

Powder Free, 6 Mills Thick, 11" Length, Blue



Size

S

M

L

XL

☐ Boxes

ADD TO ORDER

1-5 Boxes \$19.43

6 or more \$17.27

52555T2

Thickness	6 mils
Length	11"
Color	Blue
Surface	Smooth
Box Quantity	50
Additional Specifications	Powder Free Chemical-Resistant Nitrile Order 52795T216 for a compatible clear plastic dispenser (10" × 6 3/8" × 3 3/4"; holds one box of gloves). Chemical Compatibility of Gloves and Sleeves

These single-use gloves prevent cross contamination and provide the dexterity needed for intricate work. They are ambidextrous. All are made from FDA-compliant materials for food contact. Boxes of gloves fit [dispensers](#), unless noted.

Powder-free gloves are good for applications where particulate contamination is a concern.

Chemical-Resistant Nitrile—Protect your hands from incidental chemical contact such as spills and splashes. These gloves are also a good alternative for those concerned about latex allergies.

View information about the [chemical compatibility](#) of these gloves.

Figure 5. Disposable Nitrile Glove.

Outdoor Skin Protection



Sunscreen,
4 oz. Bottle

Sunscreen

			Each	
			1-3	4-Up
SPF 30 Lotion, 4-oz. Bottle	55645T8	\$6.92	\$6.29	
			Per Box	
			1-3	4-Up
SPF 30 Lotion, Single-Use Packets (50/Box)	55645T55	30.60	27.54	

Figure 6. Outdoor Skin Protection.

Face Shield with Protective Coating

Clear, Scratch-Resistant



☐

 Each

In stock

1-2 Each \$30.47

3 or more \$27.42

10865T1

ADD TO ORDER

Window Coating and Color	Scratch-resistant clear
Additional Specifications	Face Shields Standard Coatings
Related Product	Replacement Windows

The window on these face shields is coated for an added level of protection. All shields have an 8" Ht. × 14" Wd. × 0.08" Thick, polycarbonate window that's oil and grease resistant. They have ratchet adjustment to fit head sizes 6 1/2 to 8—simply turn the knob to adjust. Meet ANSI Z87.1-2003 high-impact (also known as Z87+) standards when worn with safety glasses or goggles.

Scratch-resistant coatings extend window life.

Figure 7. Face Shield with Protective Coating.

Sunglasses



Shade your eyes while working outdoors with these durable sunglasses. They have scratch-resistant polycarbonate lenses that provide 99.9% UV protection. Meet ANSI Z87.1-2003 high-impact standards (also known as Z87+). Temple-to-temple width is 5". Frame and arms are nylon with rubber tips for comfort.

Lens Color	Frame Color		Each	
			1-11	12-Up
Gray	Black	5087T2	\$3.73	\$3.40
Mirror	Black	5087T3	4.53	4.14

Figure 8. Sunglasses.

Comfort-Fit Wraparound Safety Glasses



Standard

- Scratch-resistant lenses
- Adjustable arm length
- Temple-to-temple width: 5 1/8" for standard lenses; 5 1/4" for extra-large lenses
- Polycarbonate frame; nylon arms

Cushioned temples, soft molded brow, and nonslip nose bridge make these glasses the ultimate in comfort. They have a ventilation channel between the frame and lens to increase airflow and reduce moisture buildup. Lenses have an adjustable tilt to accommodate a larger range of face sizes.



Extra Large

Indoor/outdoor lenses are clear with a slight mirror coating for varying light conditions.

Lens Color	Glasses						Replacement Lenses		
	Black Frame			Blue Frame			Each		
	Each		10-Up	Each		10-Up	Each		
	1-9	10-Up		1-9	10-Up		1-9	10-Up	
Standard Lenses									
Clear	9210T101	\$9.62	\$8.72	9210T103	\$9.62	\$8.72	9210T71	\$4.53	\$4.06
Clear, Fog Free	9210T601	10.38	9.40	9210T603	11.07	9.96	9210T76	5.35	4.79
Amber	9210T201	10.60	9.60	9210T203	10.60	9.60	9210T72	5.63	5.04
Brown	9210T301	10.60	9.60	9210T303	10.60	9.60	9210T73	5.63	5.04
Mirror	9210T401	11.60	10.55	9210T403	11.60	10.55	9210T74	6.96	6.26
Indoor/Outdoor	9210T501	11.60	10.55	9210T503	11.60	10.55	9210T75	6.96	6.26
Extra-Large Lenses									
Clear	3637T17	10.38	9.40	—	—	—	3637T44	5.21	4.67
Clear, Fog Free	3637T22	11.10	10.06	—	—	—	3637T46	6.02	5.40
Amber	3637T28	11.33	10.26	—	—	—	3637T52	6.30	5.65
Gray	3637T31	11.33	10.26	—	—	—	3637T57	6.30	5.65
Mirror	3637T33	12.28	11.16	—	—	—	3637T64	7.58	6.82
Indoor/Outdoor	3637T67	12.28	11.16	—	—	—	3637T68	7.58	6.82

Figure 9. Comfort-Fit Wraparound Safety Glasses.

3. Installation Overview

Follow this installation checklist ☒ to complete your TLS system installation.

Note: The safety of any system incorporating this equipment is the responsibility of the assembler of the system.

- ☐ Unpack the system and verify you have received the correct components and nothing is damaged. Follow Shipping and Unpacking.
- ☐ Move your TLS system to its final location. Ensure that access to the system power outlet is not obstructed (the power cord is the disconnecting device for the system). Ensure that there is at least 5 in (12.5 cm) of clear space around power supply and lamp house cooling vents. Ensure access to the back of the power supply for making System Connections (see below).
- ☐ Install the lamp. See Section 5 for X300 systems, Section 6 for Q250 systems, and Section 7 for IR Source systems.
- ☐ Review the main system components in Section 1.
- ☐ Make the system interconnections following Section 8.
- ☐ Read about how to operate the touchscreen control unit in Section 10.
- ☐ Install the SciSpec control software following the instructions in SciSpec user manual.
- ☐ Follow the initial operation instruction in Section 12 to verify that your system is operating correctly.

4. Shipping and Unpacking

4.1 Unpacking

The instrument is packaged in such a way as to minimize damage during transport. If the package is damaged or if after unpacking any signs of damage become apparent, a claim should be filed with the carrier immediately.

If the instrument must be returned, contact Sciencetech Inc. for approval prior to shipping. A full description of the reason for return should be included.

Inspect the exterior of your system for any noticeable defects. If any are present, contact Sciencetech Inc. immediately.

The lamp(s) is uninstalled before shipment (except for an IR Source). Before the lamp installation identify what type of lamp will be used.

4.2 Handling

4.2.1 Non-DUAL System

The TLS systems weighs more than 25 kg. Thus, it requires two persons to safely lift the system. The locations for lifting of the TLS is shown in Figure 10 by yellow squares. Please make sure to keep your hands away from the two gussets at each side of TLS (marked by red circle).

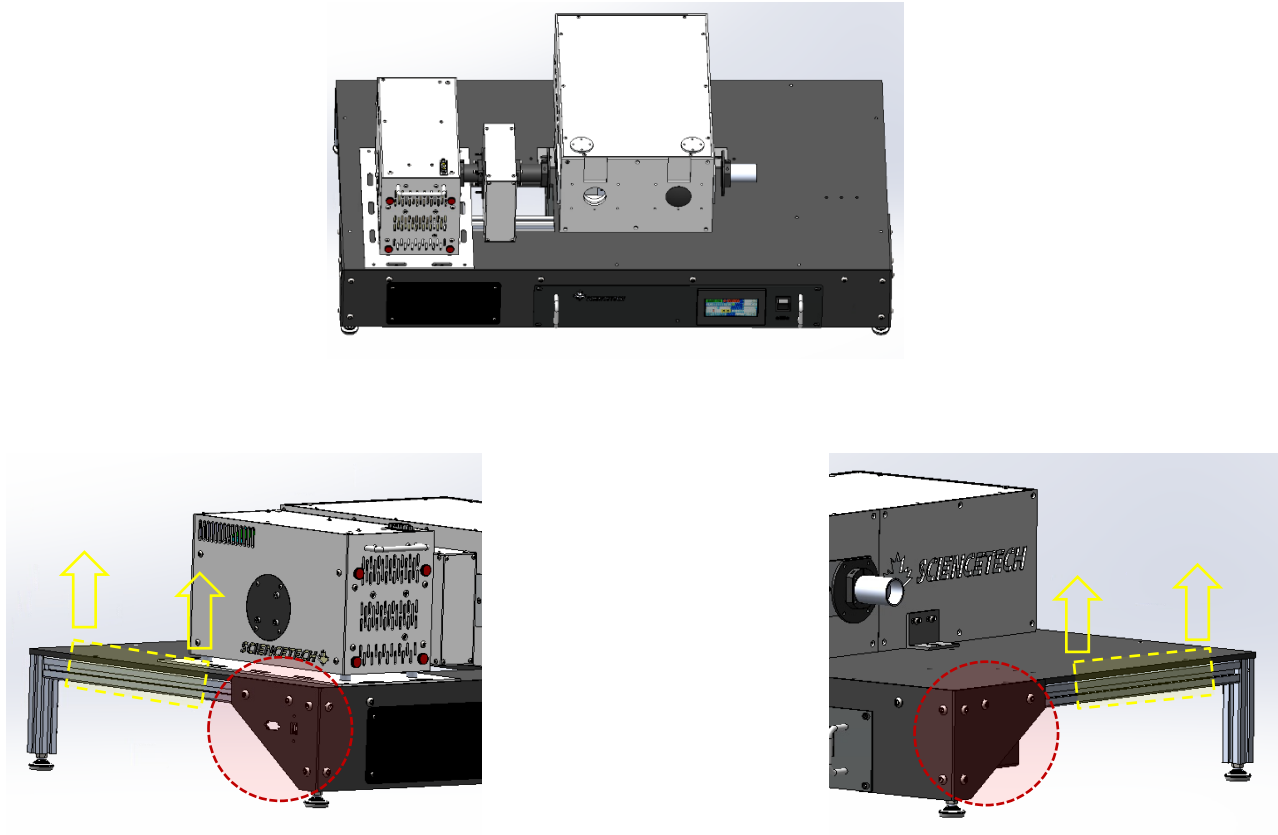


Figure 10. Lifting locations of TLS system.

4.2.2 Handling TLS-55-DUAL

TLS-55-DUAL systems weigh more than 35 kgs and has a provision for lifting with a hoist. The locations for attaching a hoist with properly rated cables are shown in Figure 11 by red circles. The hoist needs to be attached to the TLS at all 4 lift points. Your system may not be exactly as pictured below.

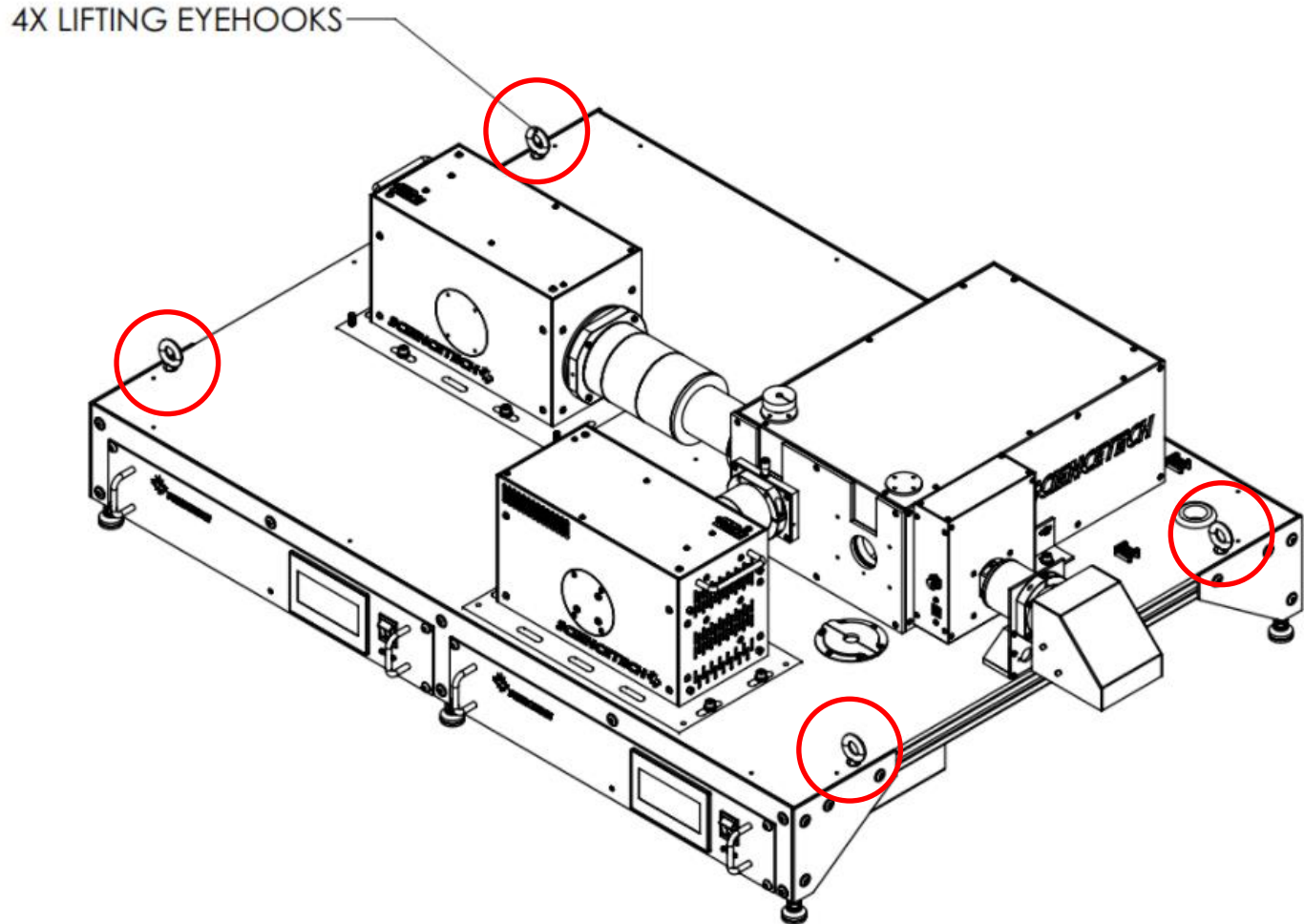
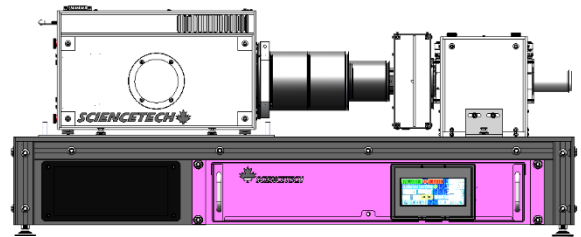
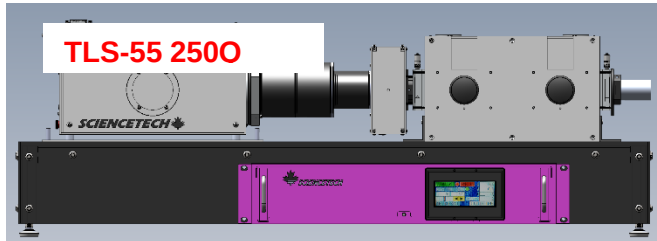


Figure 11: Lift points for a TLS-55-DUAL.

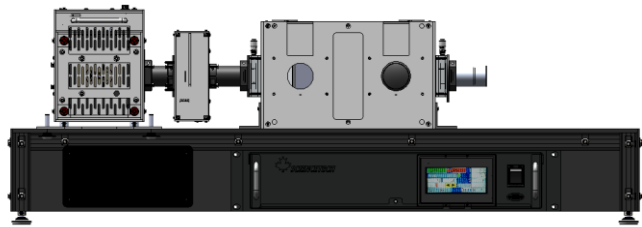
4.3 Included Components

The packing box should contain the following components:

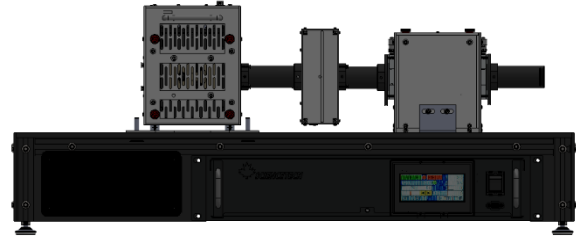
- 1) **Tunable Light Source (TLS)** with built in control unit.
 - Find your model number from your Test Report Document (TR-TLS-####)



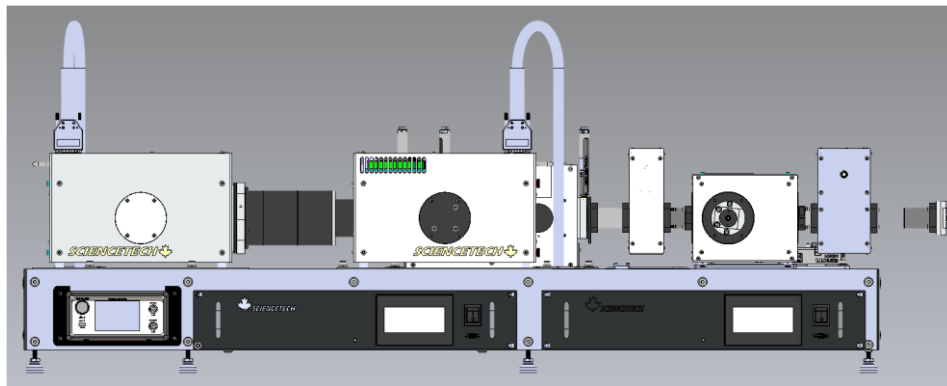
TLS-72 300X



TLS-55 2500



TLS-72-2500



TLS-55-DUAL

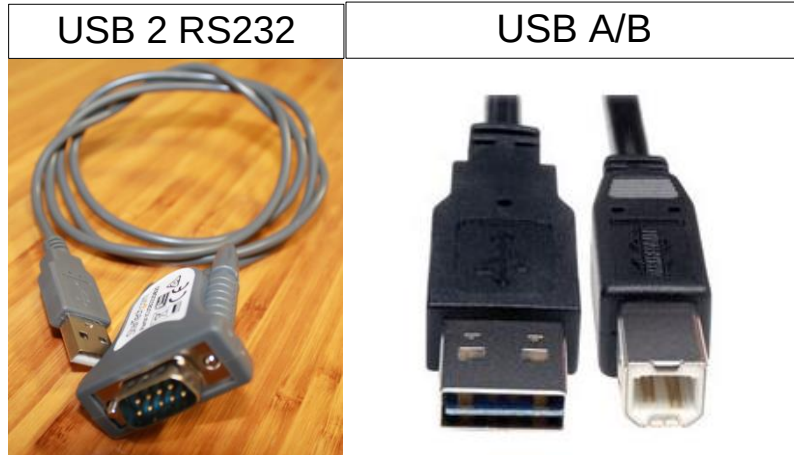
Figure 12. Different TLS systems overview.

- 2) **Power Supply Cable(s)** with IEC 320-C13 termination



Figure 13. Power supply cable with IEC 320-C13 termination.

1) Communication Cables (**USB A/B** and **USB 2 RS232**)



2) **601-CABLE**



3) TLS-PWR Cable



- 6) Xenon Arc Lamp or Quartz Tungsten Halogen lamp, depends on system model number.



Figure 14. Xenon Short Arc Lamp.



Figure 15. 250W QTH Lamp.



Figure 16: 37W IR Filament.

4.3.1 Installation of F/3 Collimating Optic Lens Tube

Please note, if you have customized the output of your TLS, it will not include the F/3 collimating optic.

The system will come with the final monochromator output optic (3" lens tube with F/3 collimating lens) disconnected and packed separately in the shipping container. To re-attach the 3" lens tube, simply insert the end with the arrow on top into the slip ring on the output slit of the monochromator. Then tighten the slip ring clamping screw with the included 3 mm Allen key. See Figure 17.

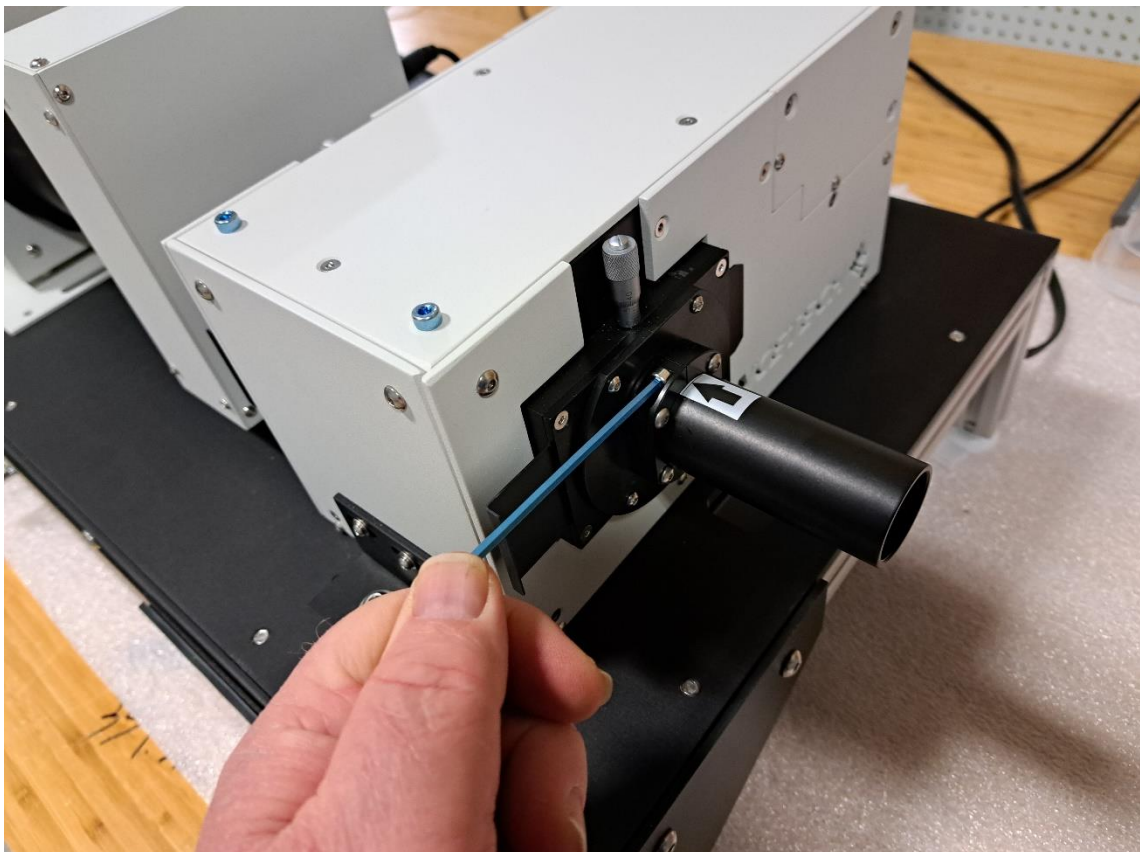


Figure 17: Attaching Collimating Optic 3" Lens Tube to Output Slit of Monochromator.

5. Instructions for Arc Lamp Systems



Before servicing the lamp housing, be sure to disconnect the electrical connections. Make certain the lamp is at room temperature. Remember to wear eye and face protection when working around arc lamps. Follow the manufacturer's general information sheets for handling and operating the bulb.

5.1 General Xenon Arc Lamp Information



Figure 18. A Xenon Arc lamp.

Lamps purchased directly from other manufacturers may not meet our specifications for operation and may cause permanent damage to the housing and to the reflector. Use of such lamps will void the warranty.



Arc lamps can be used for a MAXIMUM of 1000 hours. Exceeding 1000 hours can cause damage to the quartz bulb and may result in explosion of the lamp.

The gases inside arc lamps are under extreme pressure, especially during operation (from 10 atmospheres for large lamps, up to 30 atmospheres for small lamps). Therefore the lamp housing must be closed at all times during operation. Furthermore, precautions must be taken to ensure protection from ultra-violet radiation emissions. Special storage cases are provided to eliminate possible hazards during shipping and handling. **Safety goggles and soft cotton gloves should be worn when removing and installing lamps. Never touch**

the quartz envelope with bare hands; such handling may lead to the deterioration and premature failure. If accidentally handled, clean the lamp surface with an alcohol swab to remove any residue.



Never look directly at an operating lamp; severe eye injury will result. Wear UV protective lenses, such as welder's goggles, when working on or around lamps.

5.2 Mechanical Handling

A short arc high-pressure xenon lamp is used. The arc lamps are extremely robust. They must withstand the mechanical stresses exerted by their heavy electrodes – especially the anode, which can weigh up to 400g – and the high internal pressure of up to 30 bar (even more in low-wattage lamps). However, they are still made of glass and need to be handled accordingly; they must be protected against shock, impact and excessive force. Certain precautions must be observed when handling them.

Arc lamps are supplied in a safety cover. This protects users from possible spontaneous or induced bursting of the lamp. There is enough energy stored in the lamp bulb to send quartz splinters flying several meters across a room.

When the lamp is installed, its safety cover (if supplied) must not be removed until after it has been fitted in the lamp housing and shortly before the housing is closed. Wear protective goggles or complete face protection covering the arteries and veins of your neck, such as a transparent plastic mask. If the design of the lamp housing does not permit the lamp to be fitted together with its safety cover, it should be wrapped in strong lint-free cloth.

In some types of lamps, the safety cover acts a tool for screwing the lamp into its holder at the cathode end. **Under no circumstances must force be exerted on the bulb during installation. For example, screwing in the cathode base by holding and turning the lamp at its anode base is grossly negligent.**

The lamp may only be clamped in position at one end in order to allow for expansion and distortion of the housing. Small (short) lamps can be left free and unsupported at the other end. Larger lamps must have a soft, flexible support, which calls for a mechanical solution. It should support the lamp but allow unrestricted expansion, including expansion perpendicular to the lamp axis. Lamps may only be stored if suspended freely from their bases in their (open) safety covers. Leaving them to roll around unprotected on a desk or shelf can result in microcracks in the surface of the quartz glass, causing lamps to burst later on.

If the lamp is inadvertently operated inside its safety cover the sleeve will melt within a few seconds and the lamp will be unusable.

When removing lamps, the reverse procedure must be followed: first put the safety cover on the lamp, then remove the lamp.

Similar precautions must be observed for connecting arc lamps electrically as for handling them mechanically. All electrical connections must fulfill the criteria for high-current connections. Connecting components must be clean and offer the maximum contact area. In cases of doubt, it is better to recondition or preferably replace the

contacts rather than risk lamp failure due to corroded and overheated contacts. In most cases good electrical contact is synonymous with good thermal contact to dissipate the heat produced by the lamp.

Care must be taken to ensure that the lamp is connected with correct polarity; the positive pole of the rectifier must be connected to the base marked (+), the negative pole to the base marked (–). Incorrect polarity results in total lamp failure within a few seconds; the cathode fuses over immediately as a result of being overloaded while acting as the anode.

Arc lamps must only be held by their base. Damage to the glass may cause the lamp to break during later operation.

If the quartz bulb or the shafts should ever be inadvertently touched with bare fingers (which should never happen because unprotected lamps should only ever be handled with soft cotton gloves), the fingerprints must be removed immediately. A lint-free alcohol-moistened cloth is best for this, after which the lamp should be rubbed dry, taking care not to scratch the quartz glass surface. If fingerprints are not removed, they burn into the quartz glass surface where they act as a seed for ever-expanding recrystallization of the glass. This causes the glass to lose its strength and increases the risk of bursting.

5.3 Xenon Arc Lamp Installation

The following procedure can be used to install the xenon arc lamp into the housing:

1. Remove the four blue SHCS screws on the back of the lamp housing.

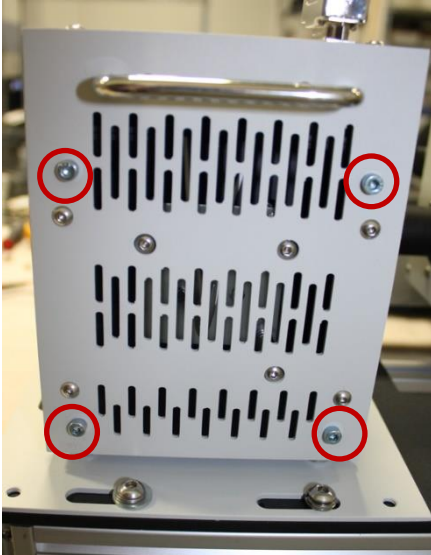


Figure 19. Blue M4x0.7 SHCS screws at the rear panel of TLS-55-X300 and TLS-72-X300.

2. Very carefully pull the back off of the lamp housing. The back of the lamp house should be pulled directly outwards. There is a puzzle piece shape attached to the lamp house top plate and must be pulled directly upwards or it will not fit through the exit opening.
3. Install the lamp by screwing the anode (+) into the brass base. Hold the lamp by the anode base as you screw the lamp into place.
4. Attached the cathode wire (black wire with round washer/lug) to the cathode end of the lamp (Figure 20). Secure the cathode wire lug to the lamp cathode with the included nut (washers not included or needed on all models).

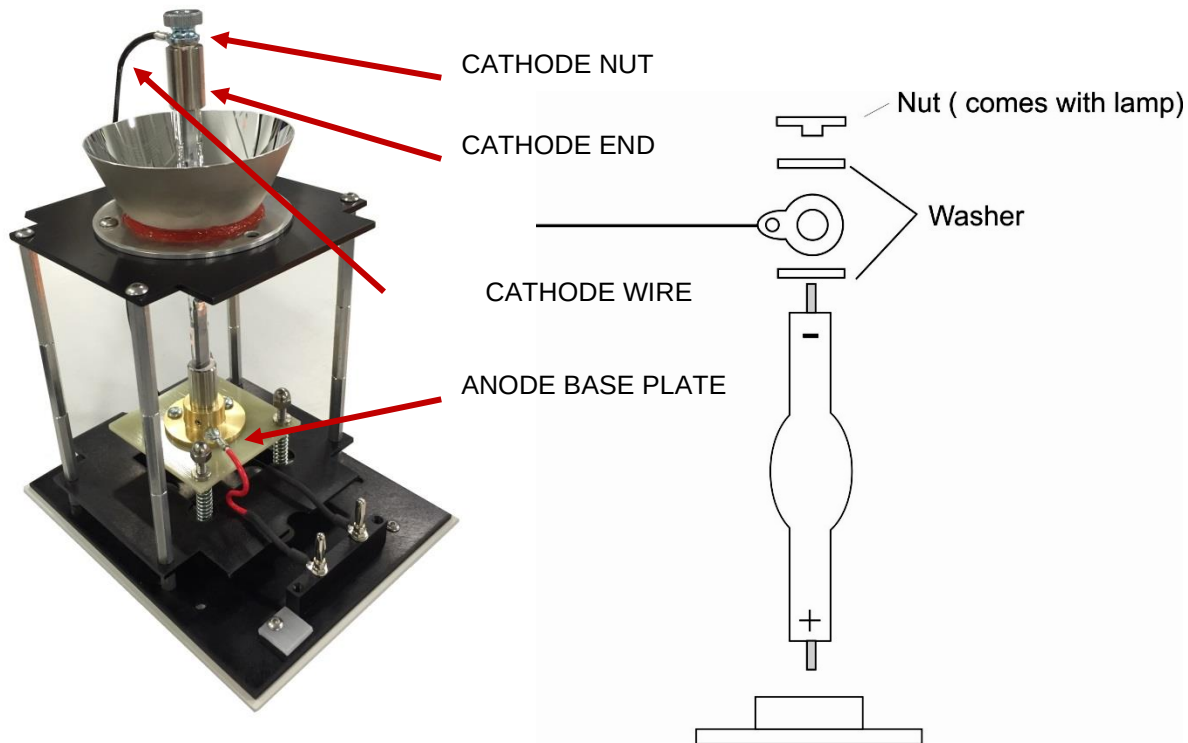


Figure 20. Installed xenon arc lamp into a LH-E lamp stage.

5. Screw the nut tight with your hand. Support the lamp cathode against the turn so that no stress is transferred to the glass.
6. Ensure PROPER POLARITY of the lamp (i.e. RED to ANODE (already wired to the brass piece) and BLACK to CATHODE).
7. CAREFULLY replace the top panel of the lamp housing. **As you insert the lamp tilt it towards the rear reflector slightly so that you can clear the internal lenses without scratching the lamp.**
8. For a new lamp installation, you may need to align the lamp. Use the following alignment procedure.



If your system is being shipped from Sciencetech with a lamp then you will not need to align the lamp, it has been done for you at the Sciencetech factory.

6. Instructions for QTH Lamp Systems

6.1 General QTH Lamp Information



Wear gloves while handling a QTH lamp. If you are removing a lamp, ensure it has had ample time to cool down after operation before handling.



Figure 21. A Standard QTH Lamp.

6.2 QTH Lamp Installation

1. Remove the top (or back depending on model) of the lamp house by removing the four blue SHCS.

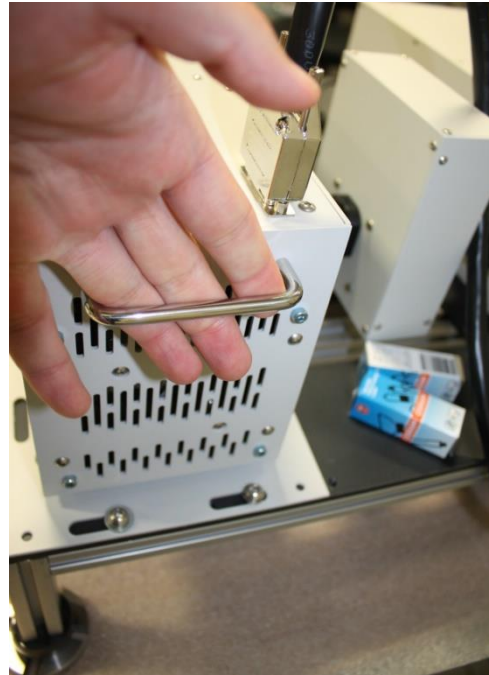
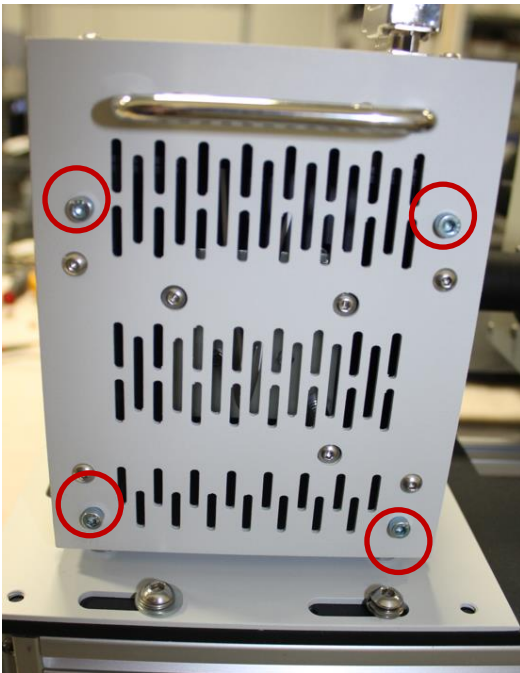


Figure 22. Removing the lamp stage form the LH-S lamp housing of TLS-55-Q250 and TLS-72-Q250.

2. Remove the QTH lamp from its packaging and handle it by the base. Slide the two prongs on the bottom of the QTH lamp into the corresponding receptacles on the lamp house stage. Slide the two prongs in until the prongs bottom out. Only a small amount of force is required to install the QTH lamp. Excessive force will damage the QTH lamp.

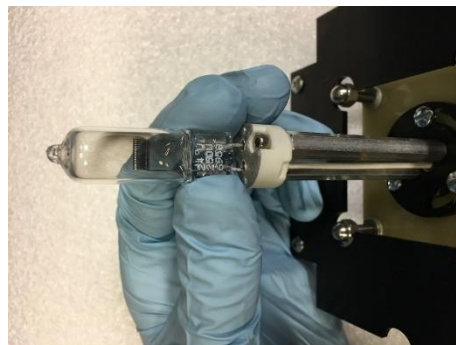


Figure 23. Inserting QTH lamp into the LH-S lamp housing.

Place the lamp base back into the lamp house and secure with the blue SHCS.

7. Instructions for IR Source Systems

7.1 General IR Source Lamp Information



Wear gloves while handling an IR Source. If you are removing a lamp, ensure it has had ample time to cool down after operation before handling. **DO NOT TOUCH THE EMITTING AREA.**

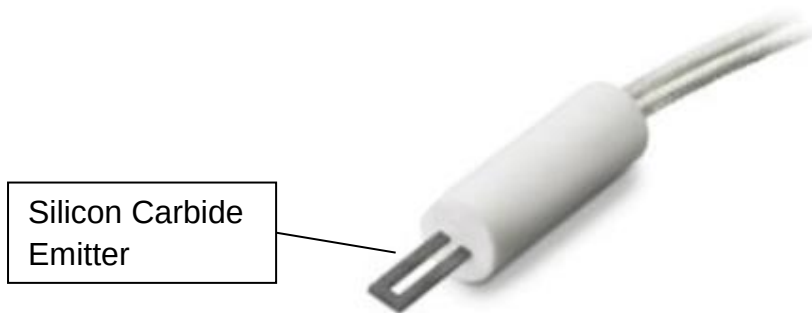


Figure 24. An IR Source.

7.2 IR Source Installation

The following procedure can be used to install the IR Source into the housing:

1. Remove the four blue SHCS screws on the back of the lamp housing.

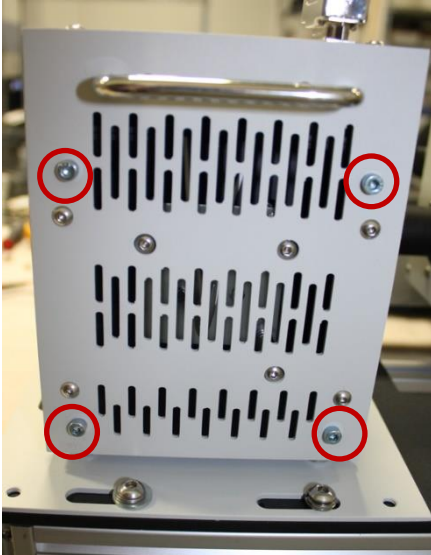
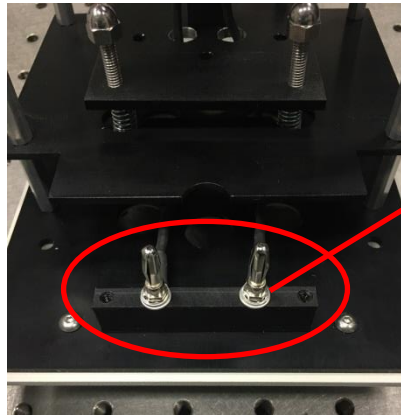
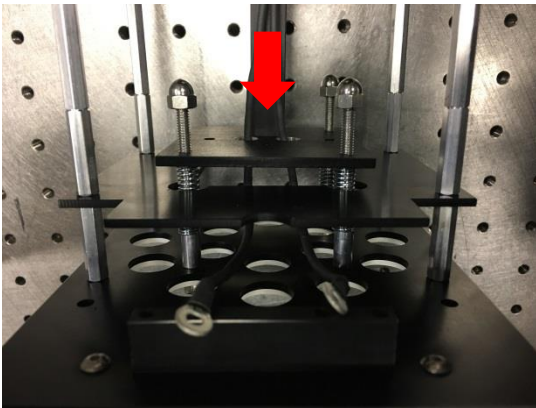
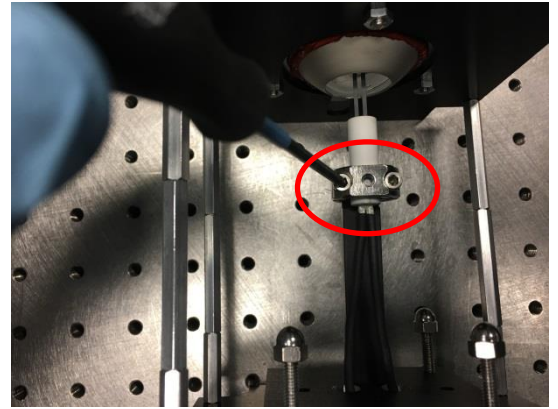
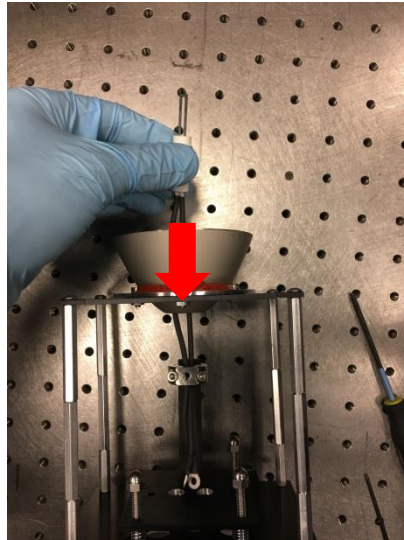
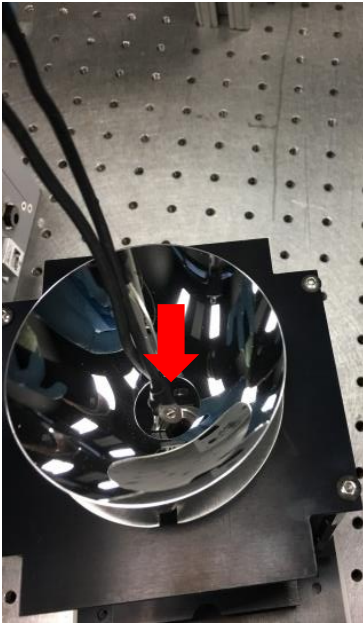


Figure 25. Blue M4x0.7 SHCS screws at the rear panel of the lamp house.



2. Very carefully pull the back off of the lamp housing. The back of the lamp house should be pulled directly outwards. There is a puzzle piece shape attached to the lamp house top plate and must be pulled directly upwards or it will not fit through the exit opening.
3. While wearing gloves, carefully place the wires and IR source through the opening of the reflector. Ensure that the wires, connectors, and IR source do not hit the reflector surface. Only handle the IR source by the white base. Ensure the wires also go through the slip ring opening as shown below.
4. Slide the IR source into the slip ring and tighten down the slip ring with a 7/64" Allen key.
5. Ensure the wires have gone through the two holes in the plate the IR source is mounted to.
6. Remove the two banana jack connections.
7. Slide the eyelets onto the banana jacks and re-attach the banana jacks to the lamp stage.



IR Source eyelets
connected to the
banana jacks.



If your system is being shipped from Sciencetech with a lamp then you will not need to align the lamp, it has been done for you at the Sciencetech factory.

8. System Connections

8.1 Interconnection Table

Table 2. Interconnection table.

Label	Description	Connector (Gender)	System	Interconnection
7W2	Lamp power input to lamp house	7W2 (M)	LH	601-CABLE
PWR	Power Input (100/230 VAC)	IEC 320-C14 (M)	601 Supply	All IEC 320-C13 (F)
RS232	DB9	DB9 (F)	601 Supply	USB to RS232
SH	Accessory Cable	6pin Mode (M)	601 Supply	TLS-PWR
7W2	Power and communication between 601 and lamp house	7W2 (F)	601 Supply	601-CABLE
24VDC	24 VDC input	Barrel (F)	Filter Wheel	TLS-PWR
USB FW	USB in to filter wheel controller	USB B	Filter Wheel	USB cable A-B, 6ft
24VDC	24 VDC input	Barrel (F)	Monochromator	TLS-PWR
USB M	USB in to monochromator controller	USB B	Monochromator	USB cable A-B, 6ft

8.2 System Connection Instructions

8.2.1 Standard TLS

1. Connect the **TLS-PWR** cable to **SH** on the power supply and **24VDC** on the monochromator and filter wheel.
2. Connect the **USB A/B** cables from **USB M** and **USB FW** to the **control computer**.
3. Connect the **USB to RS232** cable from **RS232** to the **control computer**.
4. Connect **601-CABLE** between the **7W2** connections on the power supply and lamp house.
5. Ensure the power supply is off. Connect the **provided line cord** with IEC 320-C13 termination to **PWR**. This cord is the disconnecting device for the TLS system. Do not position the equipment so that it is difficult to unplug this cord (i.e., operate the disconnecting device).
6. For any custom accessories follow the next sections figure and table for custom component hookup instructions.

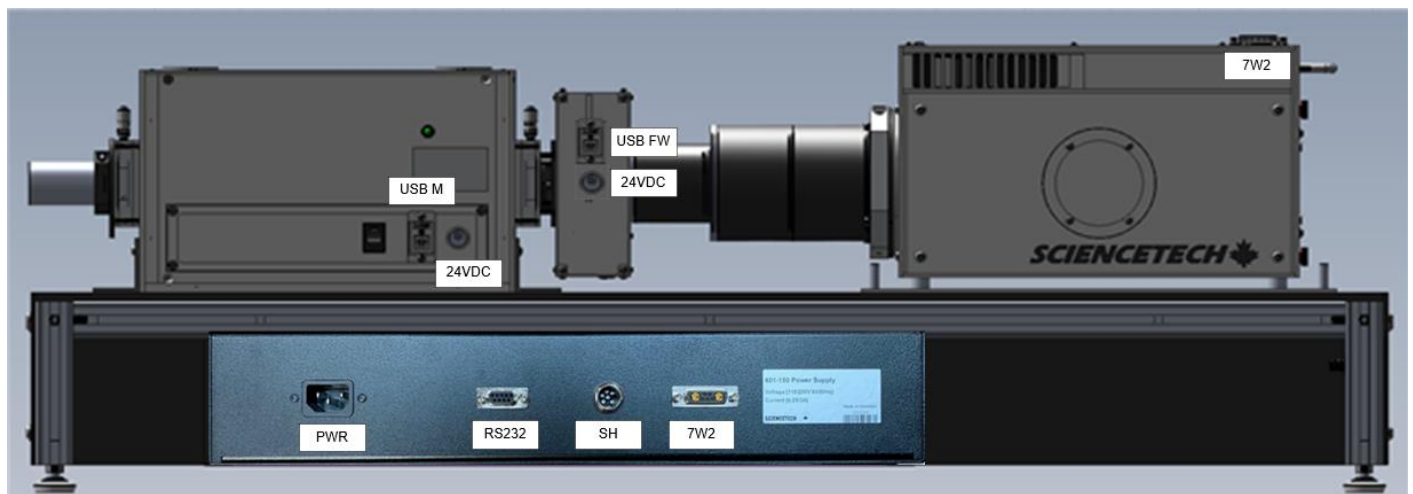


Figure 27: Connection location figure.

8.2.2 TLS-55-DUAL

1. Connect the **7W2 cable** to each lamp house based on the label on the cable (i.e 37W IR is for the LH-E-37IR and QTH 250W is for the LH-S-Q250).
2. Connect the **USB cables** tapped to the top of the TLS to **USB M** and **USB FW** and **24 VDC** (x2) on the Monochromator and Filter Wheel.
3. Ensure the power supply is off. Connect the **provided line cord** with IEC 320-C13 termination to **power entry on the rear of the TLS**. This cord is the disconnecting device for the TLS system. Do not position the equipment so that it is difficult to unplug this cord (i.e., operate the disconnecting device).
4. Connect the **USB A/B** cable from the rear of the TLS to the **control computer**.

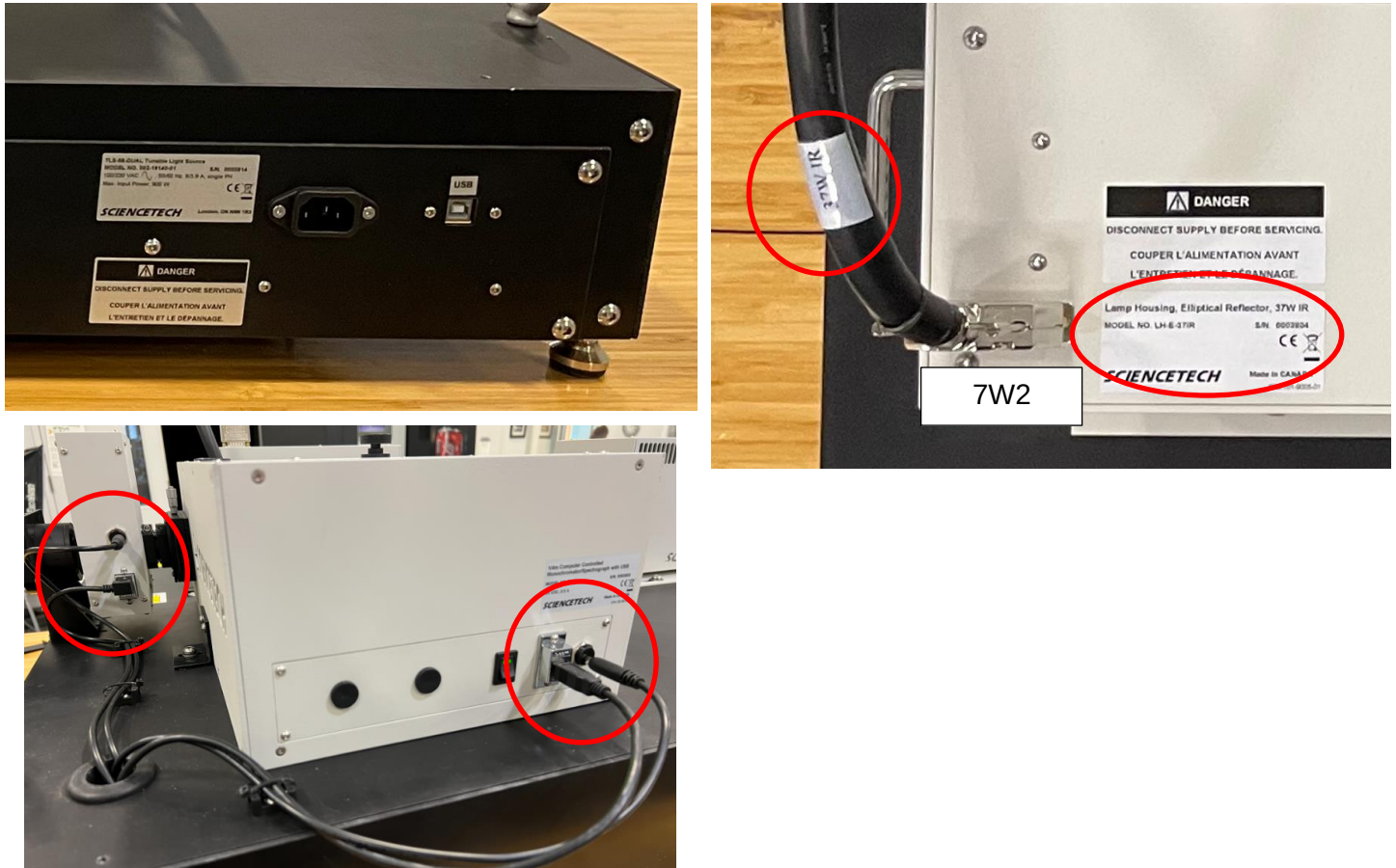


Figure 28: Power entry and USB on the rear of a TLS-55-DUAL. 7W2 cable connection example. Monochromator and filter wheel connections.

9. General Dimensions

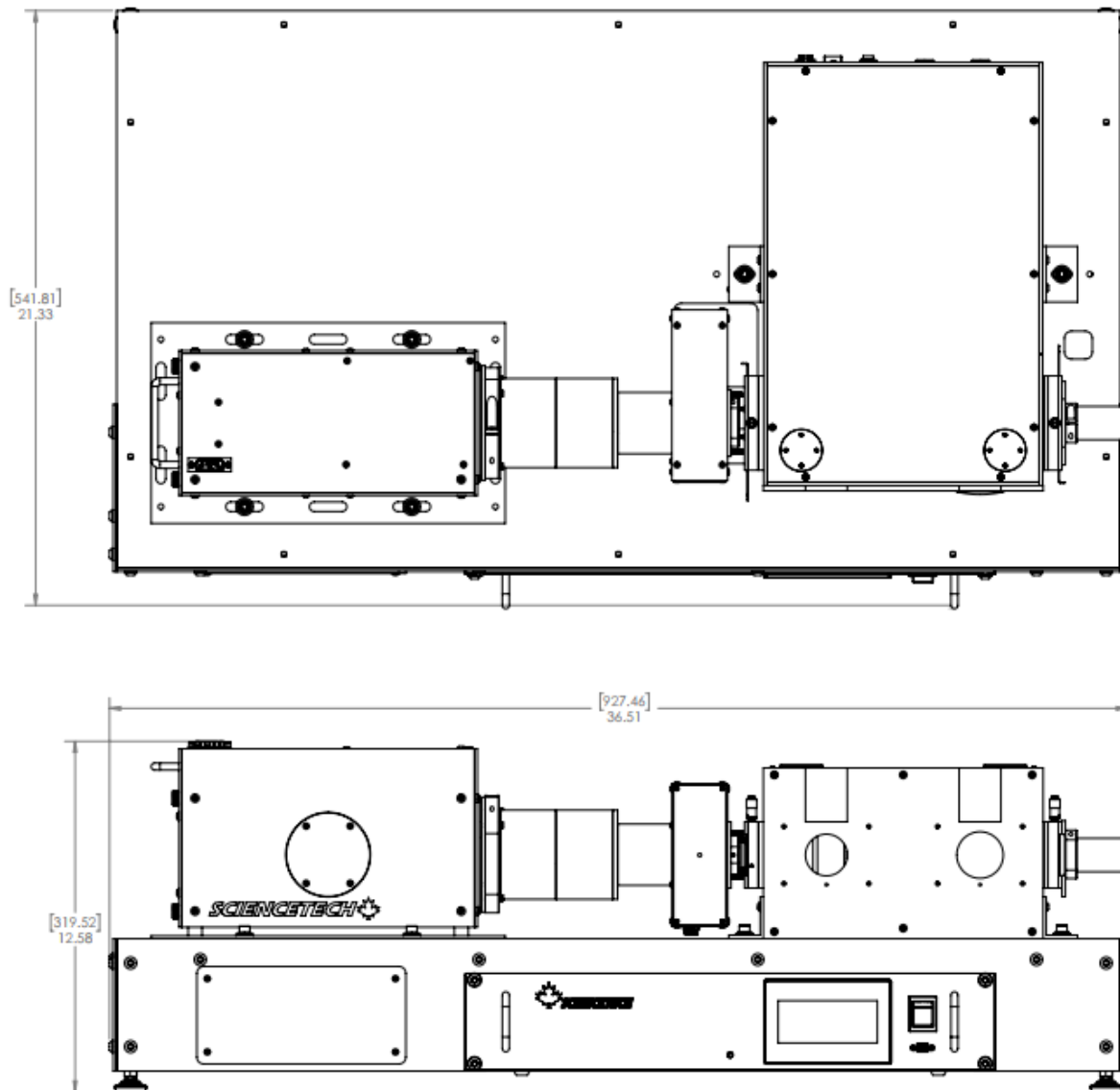


Figure 29. Dimensions of TLS-55-X300.

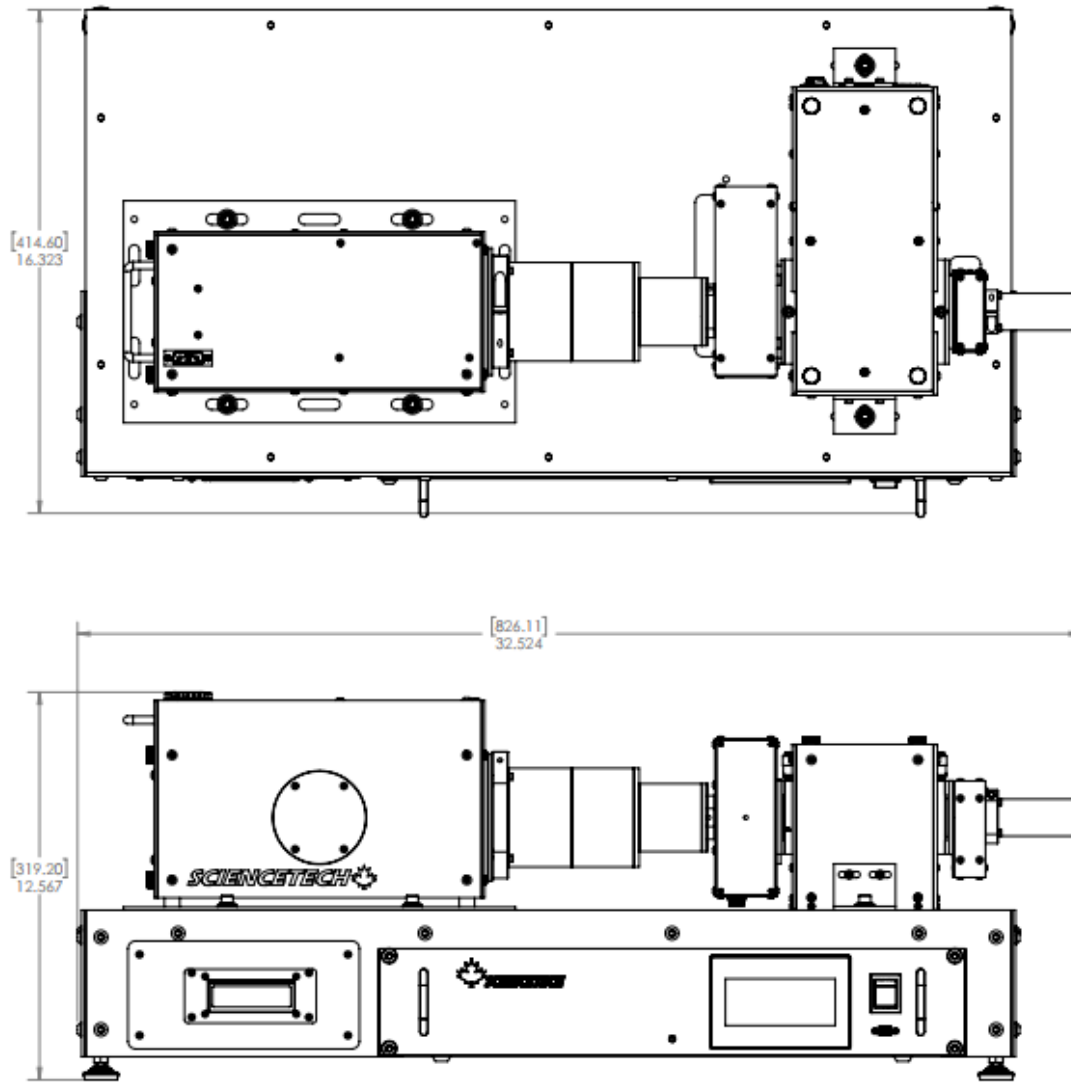


Figure 30. Dimensions of TLS-72-X300.

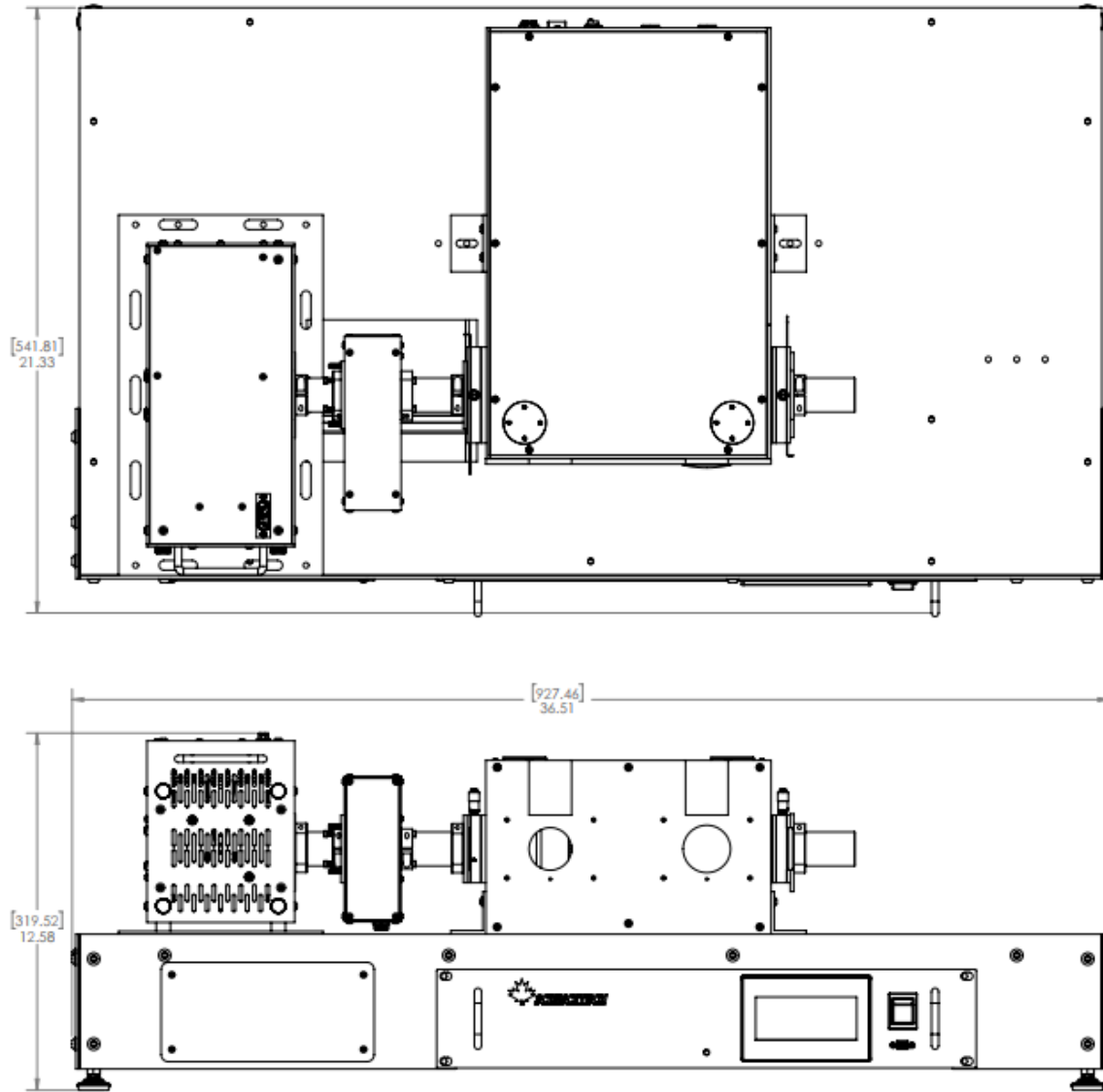


Figure 31. Dimensions of TLS-55-Q250.

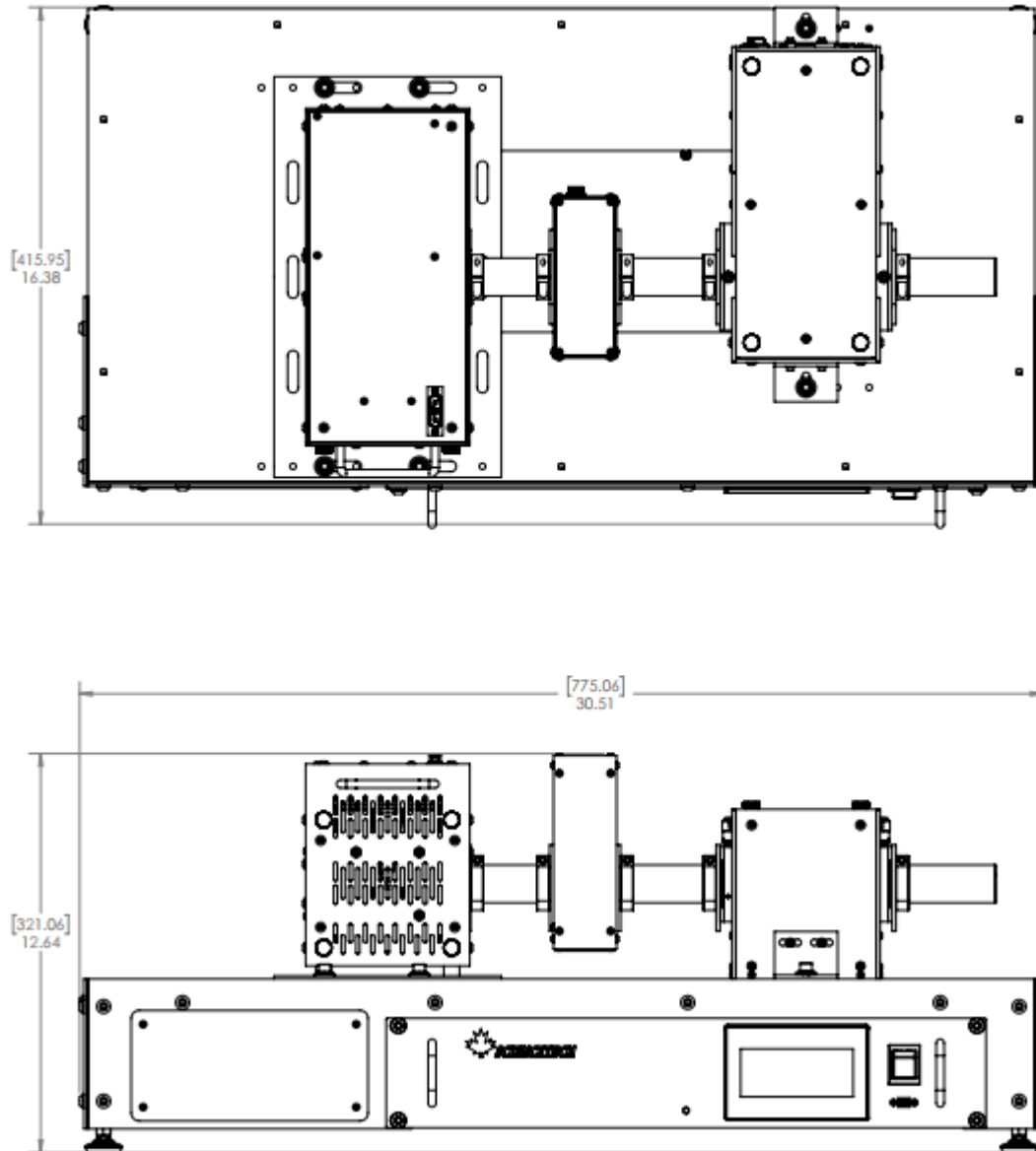


Figure 32. Dimensions of TLS-72-Q250.

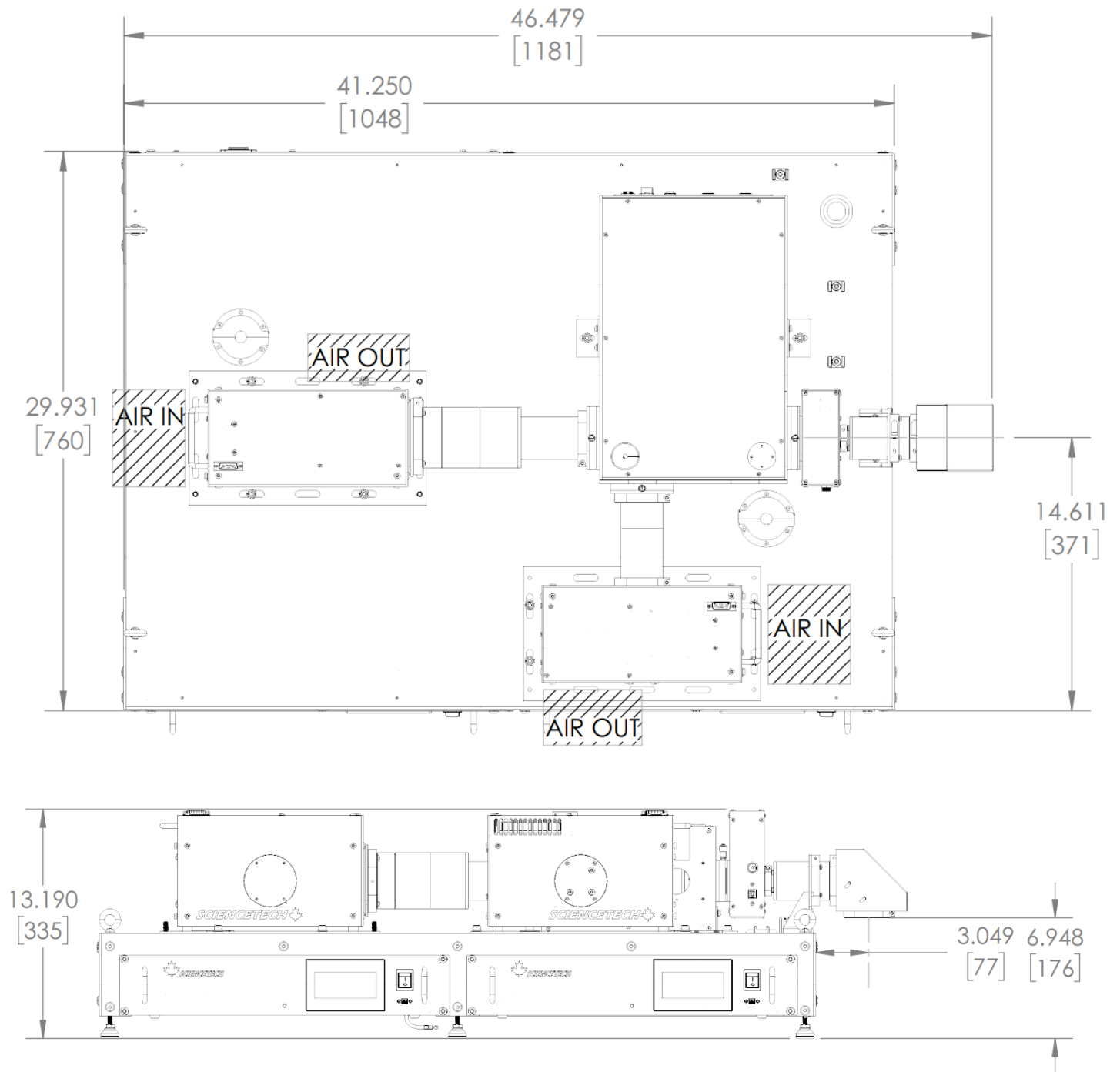


Figure 33: TLS-55-DUAL dimensions.

10. Power Supply Description and Operation



If this equipment is used in a manner not specified by the Sciencetech, the protection provided by the equipment may be impaired.

Please note that the PLC firmware and the operation procedure for 602 series power supplies (for QTH lamps used in TLS-72-Q250 and TLS-55-Q250) are different than 601 series power supplies.

10.1 601 Series Power Supply Overview and Connections

All connections to these power supplies are on the back panel, as shown below.

10.1.1 Back Panel Connections

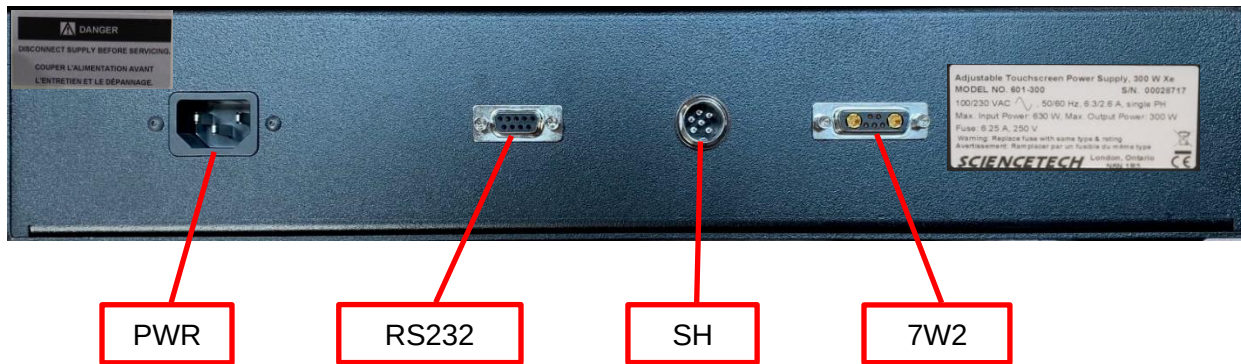


Figure 34. Rear of 601 series power supply with connections labelled.

Table 3. Label acronyms, descriptions and connector types explained.

LABEL	DESCRIPTION	601 SERIES	CONNECTOR
PWR	Power Input (120V OR 220V AC)	Y	IEC 320-C14
RS232	Serial input for Computer Control. The other end of this cable is a USB for interfacing with the Power Supply Control Software.	Y	DB9 Female
SH	Shutter Control and Accessory Fan Power	Y	6-pin Mode
7W2	Power to Lamp and Cooling Fans	Y	7W2

To connect the power supply for use, plug the provided 7W2 cable into the 7W2 connector on the lamp house and power supply unit. Plug the line cord into PWR (100-130VAC OR 200-230VAC depending on model of supply) and to the appropriate wall outlet. Plug the RS232 into the RS232 connector of the power supply and the other end to a USB port on your computer. Note that the line cord is the disconnecting device for this equipment. Do not position the equipment so that it is difficult to unplug the line cord from the outlet. Do not replace the line cord with an inadequately rated cord (requirements are in Table 5).

Optional: If you purchased an electromechanical shutter. Connect 2 or 6 pin shutter cable to SH. Shutter is usually located inside of a lamp house or solar simulator.

10.1.2 Touchscreen Types

Sciencetech uses two types of touchscreens in the 601 series power supply: Type I and Type M. The touchscreens can most easily be distinguished by the size of the touchscreen. The two types are shown below in Figure 35.

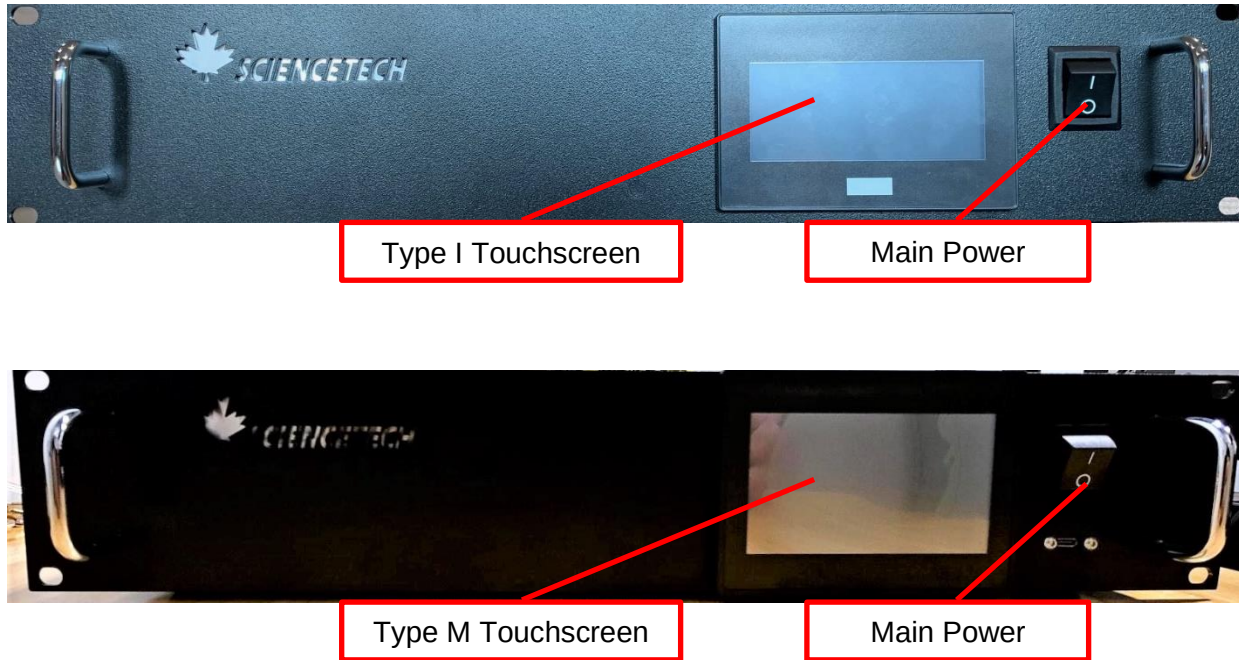


Figure 35. 601 Front Panels showing Type I (top) and Type M (bottom) touchscreens.

The user instructions for **type I** touchscreens are located in Section 10.2 The user instructions for **type M** touchscreens are located in Section 10.3.

10.2 Control Unit Operation for Arc Lamp Systems (601- Power Supply Type I)

Xenon arc lamp-based systems used 601 series power supplies.

10.2.1 Power Supply Start-up



Figure 36: Opening screen (Type I PLC)

After making all connections outlined in the above setup procedure, turn on the power supply using the main power switch on the front of the power supply. You should hear the fans in the power supply activate. If this does not happen double check that all connections have been made correctly. Once the power supply's fans have started, press OK on the starting screen.

10.2.2 Main Screen



Figure 37: Main menu (Type I PLC)

LAMP ON: Turns on arc lamp

LAMP OFF: Turns off arc lamp

V: Displays present lamp voltage

I: Displays present lamp current

P: Displays present lamp power

INDICATOR LIGHT: If the light is RED, the lamp is off. If the light is GREEN the lamp is on.

T: Displays the elapsed time in minutes since the lamp was turned on.

Fans On: Turns on lamp housing fans. Always turn on the lamp housing fans before igniting the lamp.

Fans Off: Turns off lamp housing fans. Always let the lamp housing fans run for a few minutes after turning off the lamp.

FANS OFF: Displays the present state of the lamp housing fans.

LOG: Opens the log screen with lamp and power supply run time information.

87.9 [BLUE BOX]: If the lamp is running, this box displays the present lamp current percentage. In the example above if the lamp were on it would be operating at 87.9% of its maximum current.

88.0 [YELLOW BOX]: To set a new current percentage for the lamp to operate at touch this box. A popup window will appear and allow you to type in a new value to operate the lamp at. Once complete press enter on the popup window. The change will not take effect until the green SET button is pressed.

SET: Press this button to apply the current percentage value in the YELLOW BOX to the arc lamp. This change will be reflected in the BLUE BOX that displays the current percentage the lamp is operating at. In this example the lamp would be set to 88.0% if the green SET button were pressed.

SHUTTER: Brings up the shutter control screen

10.2.3 Starting the Lamp

Ensure all connections have been made according to the instructions in this manual. Press the **Fans On** button to turn on the fans in the lamp housing. Ensure the lamp housing fans have started. Running the lamp without turning on the lamp housing fans will result in overheating. The **FANS ON/OFF** display should now show that the lamp housing fans are on. You may now press the **LAMP ON** button to ignite the lamp. The power supply has a preset ignition voltage that will generate less than the max current for your bulb. You can set a customized power level by touching the **YELLOW BOX** above the green **SET** button. A window will appear with numbers so that you may choose the percentage of total power you would like to supply. Hitting the green **SET** button will set the power supply to the value you entered. Adjusting the number displayed does nothing until the green **SET** button is pressed.

To turn off the lamp simply press the **LAMP OFF** button. A popup window will warn that the fans will continue to operate for 1-5 minutes before shutting off.

The minimum output current percentage that the power supply should be set to is 75% (the power supply may accept values below this; however, the lamp might not work properly), and the maximum value is 105%. Running the lamp outside of this range can cause damage to the lamp, however this should not be possible as all 600- series power supplies are configured so that the lamp cannot be operated outside of the safe operational current range.

10.2.4 Log Screen



Figure 38: Log screen (Type I PLC)

Access the log screen by pressing the **LOG** button on the main screen or the shutter screen.

- TOTAL:** Displays the total amount of time the power supply has run
- LAMP:** Displays the total time the current lamp has run
- LAMP STARTS:** Displays the total number of starts on the current lamp
- RESET LAMP:** Resets the total lamp time and number of lamp starts when a new lamp is installed.
- STOP LAMP:** Turns off the lamp
- MAIN:** Returns to the main screen
- SHUTTER:** Moves to the shutter screen

10.2.5 Shutter Screen

Access the shutter screen by pressing the **SHUTTER** button on the main screen or the log screen.



Figure 39: Shutter screen (Type I PLC)

OPEN/CLOSE: Manually opens and closes the shutter

Shutter Icon: Displays the present state of the shutter

OPEN/CLOSE TIME: Open (left) and close (right) time, in seconds, for a one-shot exposure or loop operation.

EXPOSE: Performs a one-shot exposure by opening the shutter for the specified open time.

START/STOP LOOP: Starts or stops the exposure loop. While the loop is running, the shutter will open and close for the times specified by **OPEN/CLOSE Time**.

STOP LAMP: Turns off the lamp

MAIN: Returns to the main screen

RUNTIME: Returns to the log screen

HELP: Shows the help screen for shutter operation

10.3 Control Unit Operation for Arc Lamp Systems (601- Power Supply Type M)

10.3.1 Power Supply Start-up

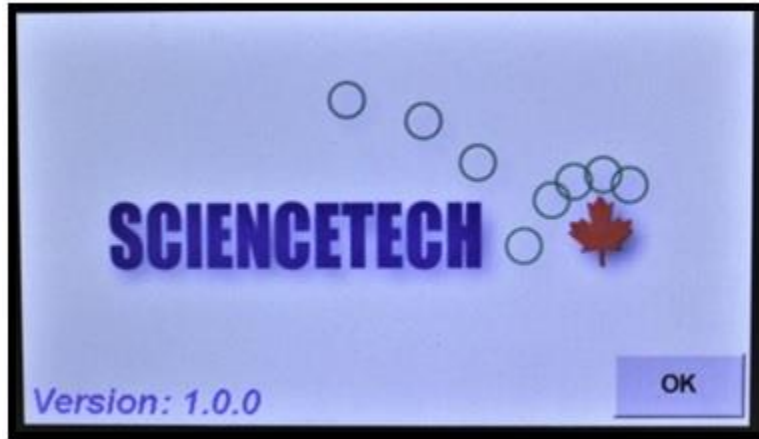


Figure 40: Opening screen (Type M PLC)

After making all connections outlined in the above setup procedure, turn on the power supply using the main power switch on the front of the power supply. You should hear the fans in the power supply activate. If this does not happen double check that all connections have been made correctly. Once the power supply's fans have started, press OK on the starting screen.

10.3.2 Main Screen



Figure 41: Main screen (Type M PLC)

Lamp ON: Turns on arc lamp

Lamp OFF: Turns off arc lamp

V: Displays present lamp voltage

I: Displays present lamp current

P: Displays present lamp power

Lamp Time: Displays time, in hours and minutes, since the lamp was turned on

LAMP ON/LAMP OFF: Indicates whether the lamp is on or off

Fan ON: Turns on lamp housing fans. Always turn on the lamp housing fans before igniting the lamp.

Fan OFF: Turns off lamp housing fans. Always let the lamp housing fans run for a few minutes after turning off the lamp.

FANS OFF/Running: Displays the present state of the lamp housing fans.

80.0 [BLUE BOX]: If the lamp is running, this box displays the present lamp current percentage. In the example above the lamp is operating at 80.0% of its maximum current.

70.0 [YELLOW BOX]: To set a new current percentage for the lamp to operate at press this box. A popup window will appear and allow you to type in a new value to operate the lamp at. Once complete, press enter on the popup window. The change will not take effect until the green SET button is pressed.

Set: Press this button to apply the current percentage value in the YELLOW BOX to the arc lamp. This change will be reflected in the BLUE BOX that displays the current percentage the lamp is operating at. In this example the lamp would be set to 70.0% if the green SET button were pressed.

Menu: Opens the menu screen

10.3.3 Starting the Lamp

Ensure all connections have been made according to the instructions in this manual. Press the **Fans On** button to turn on the fans in the lamp housing. Ensure the lamp housing fans have started. Running the lamp without turning on the lamp housing fans will result in overheating. The **FANS OFF/Running** display should now show that the lamp housing fans are on. You may now press the **Lamp ON** button to ignite the lamp. The power supply has a preset ignition voltage that will generate less than the max current for your bulb. You can set a customized power level by touching the **YELLOW BOX** above the green **Set** button. A window will appear with numbers so that you may choose the percentage of total power you would like to supply. Hitting the green **Set** button will set the power supply to the value you entered. Adjusting the number displayed does nothing until the green **Set** button is pressed.

To turn off the lamp simply press the **Lamp OFF** button. A popup window will warn that the fans will continue to operate for 1-5 minutes before shutting off.

The minimum output current percentage that the power supply should be set to is 75% (the power supply may accept values below this; however, the lamp might not work properly), and the maximum value is 105%. Running the lamp outside of this range can cause damage to the lamp, however this should not be possible as all 600- series power supplies are configured so that the lamp cannot be operated outside of the safe operational current range.

10.3.4 Menu Screen



Figure 42: Menu screen (Type M PLC)

Access the menu screen by clicking the **Menu** button on the main screen.

Statistics: Opens a screen displaying lamp run statistics

Feedback: Opens a screen for configuring lamp power feedback. This is an optional capability. If it is not installed the message will be "Not Installed," and if it is installed the message will be "Open."

Shutter: Opens a screen for configuring the automatic shutter. This is an optional capability. If it is not installed the message will be "Not Installed," and if it is installed the message will be "Open."

10.3.5 Statistics Screen

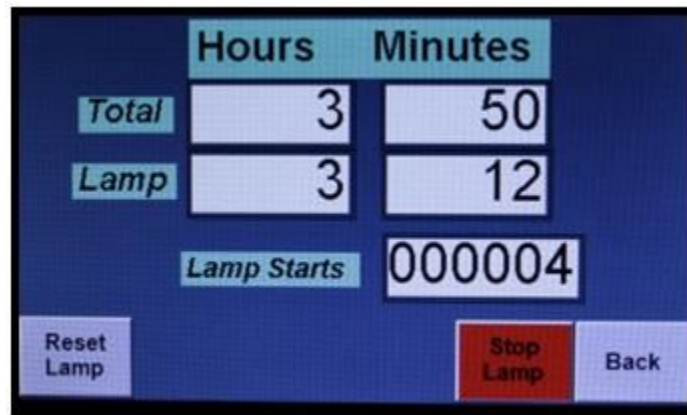


Figure 43: Statistics screen (Type M PLC)

Access the Statistics screen by clicking the Statistics button on the Menu screen

Total: Displays the total amount of time the power supply has run

Lamp: Displays the total time the current lamp has run

Lamp Starts: Displays the total number of starts on the current lamp

Reset Lamp: Resets the total lamp time and number of lamp starts when a new lamp is installed

Stop Lamp: Turns off the lamp

Back: Returns to the main menu

10.3.6 Shutter Screen



Figure 44: Shutter screen (Type M PLC)

Access the shutter screen by pressing the **Shutter** button on the menu screen.

Open: Opens the shutter

Close: Closes the shutter

Open/Close Time: Open (left) and close (right) time, in seconds, for a one-shot exposure or loop operation

Expose: Performs a one-shot exposure by opening the shutter for the specified open time.

Start/Stop Loop: Starts or stops the exposure loop. While the loop is running, the shutter will open and close for the times specified by **Open/Close Time**.

Stop Lamp: Turns off the lamp

Help: Shows the help screen for shutter operation

Main: Returns to the main screen

Run Time: Returns to the statistics screen.

10.4 Control Unit Operation for QTH and IR Source Systems (602 Power Supply)

602-series power supplies are designed to work with Sciencetech XLH-series lamp houses using QTH lamps or IR emitters. They include a USB to RS232 serial adaptor to allow remote control of the power supply, a lamp life timer, and a touchscreen interface to control the Sciencetech XLH lamp houses.

After making all connections outlined in the above setup procedure, turn on the power supply using the main power switch on the front of the power supply. You should hear the fans in the power supply activate. If this does not happen double check that all connections have been made correctly. Once the power supply's fans have started, press OK on the starting screen.



Figure 45: Start screen (602 series power supply)

10.4.1 Main Screen

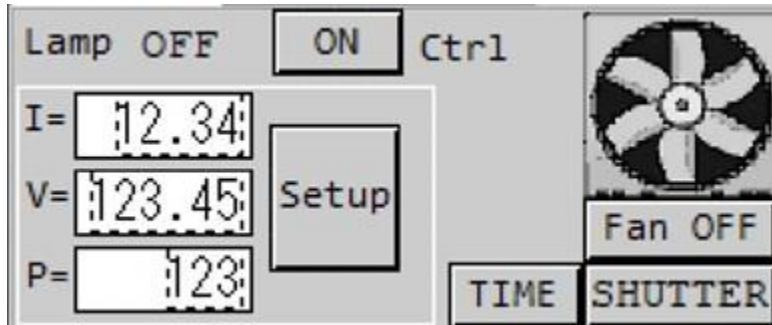


Figure 46: Main screen (602 series power supply)

ON/OFF: Turns the lamp on or off. The lamp cannot be turned on unless the lamp house fans are running.

Lamp ON/OFF: Indicates whether the lamp is on or off

I: Displays present lamp current

V: Displays present lamp voltage

P: Displays present lamp power

Setup: Brings up the setup screen which is used to set the lamp power

Fan ON/OFF: Turns the lamp house fans on or off. The lamp house fans cannot be turned off if the lamp is running.

TIME: Brings up the time screen which provides system and lamp run time and lamp start data

SHUTTER: Brings up the shutter screen which provides controls for the operation of an optional automated shutter.

10.4.2 Setup Screen

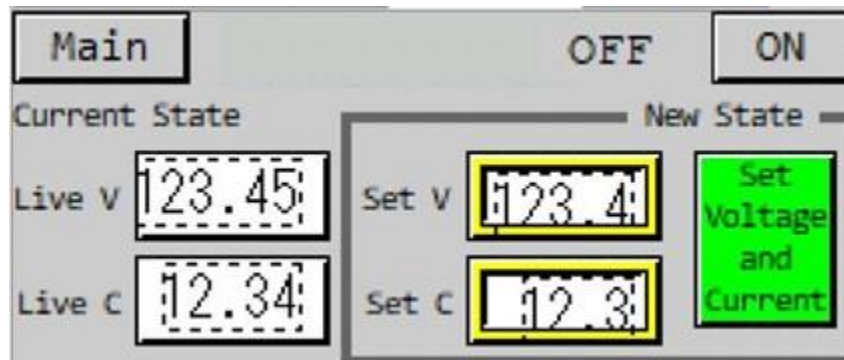


Figure 47: Setup screen (602 series power supply)

Main: Returns to the main screen

ON/OFF (Indicator): Indicates whether the lamp is on or off

ON/OFF (Control): Turns the lamp on or off. The lamp cannot be turned on unless the lamp house fans are running.

Live V: Displays the present lamp voltage

Live C: Displays the present lamp current

Set V: Entry for new lamp voltage. When pressed, a popup window will appear allowing you to type in a new value for the lamp voltage. Once complete, press enter on the popup window. The change will not take effect until the green **Set Voltage and Current** button is pressed.

Set C: Entry for new lamp current. When pressed, a popup window will appear allowing you to type in a new value for the lamp current. Once complete, press enter on the popup window. The change will not take effect until the green **Set Voltage and Current** button is pressed.

Set Voltage and Current: Changes the lamp power supply voltage and current setting to the values contained in **Set V** and **Set C**

Note that the fan will turn off after approximately one minute of operation if the lamp is not on. Thus, if the setup time takes longer than that, the fan will turn off and must be restarted before starting the lamp. To do so, simply press **Fan ON** on the main screen again. Finally, after turning off the lamp the fan will automatically shut down after approximately one minute.

10.4.3 Starting the Lamp

Press the **Setup** button and set the desired voltage and current values. The power supply will not exceed either of these values during operation.

If you want to set a particular lamp voltage, enter that value in **Set V**, and enter a value into **Set C** that is HIGHER than the expected lamp current. This will cause the power supply to operate at the desired voltage while supplying a current less than that set in **Set C**. The power supply is thus limiting on voltage.

If you want to set a particular lamp current, enter that value in **Set C**, and enter a value into **Set V** that is HIGHER than the expected lamp voltage. This will cause the power supply to operate at the desired current while supplying a voltage less than that set in **Set V**. The power supply is thus limiting on current.

When the desired voltage and current values have been entered, press the green **Set Voltage and Current** button.

When finished, press **Main** to return to the main screen. If the fans are not running, turn them on. Then press **ON** to turn on the lamp. You should see the live voltage, current, and power levels displayed on the main screen.

10.4.4 Time Screen

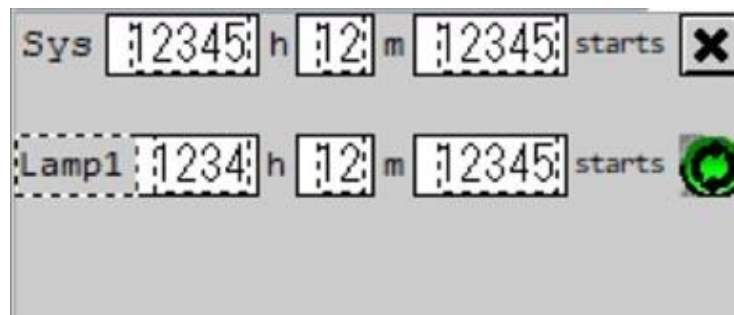


Figure 48: Time screen (602 series power supply)

Sys (__ h, __ m, __ starts): Displays the total number of hours and minutes that the system has run and the total number of lamp starts

Lamp1 (__ h, __ m, __ starts): Displays the total number of hours and minutes that the current lamp has run and the total number of times the current lamp has started

Green Button: Resets the **Lamp1** values to zero. Press this button when a new lamp is installed.

X: Returns to the main screen

10.4.5 Shutter Screen

To operate the shutter (optional accessory), simply press SHUTTER on the main screen followed by the **OPEN/CLOSE** button.



Figure 49: Shutter screen of 602 series power supply

Main: Returns to the main screen

CLOSE/OPEN: Closes or opens the shutter. The graphic displays the current state of the shutter.

11. Software Installation

Each TLS system comes with a version of SciSpec software which is used to control the system and facilitate wavelength selection and scanning. Follow the installation instructions SciSpec user manual.

12. General System Operation

1. Make sure the SciSpec software has been installed.
2. With the software closed, follow the cabling connections shown in Section 0.
3. Ignite the lamp following Section 10.
4. Set the lamps to the desired power, typically the maximum.
5. Open SciSpec.



NOTE: The Power Supply(ies) must be turned on in order for the monochromator and filter wheel to receive power.

12.1 Setting System Bandpass

Set the desired slit widths on the monochromator by using the manual micrometer adjustment.

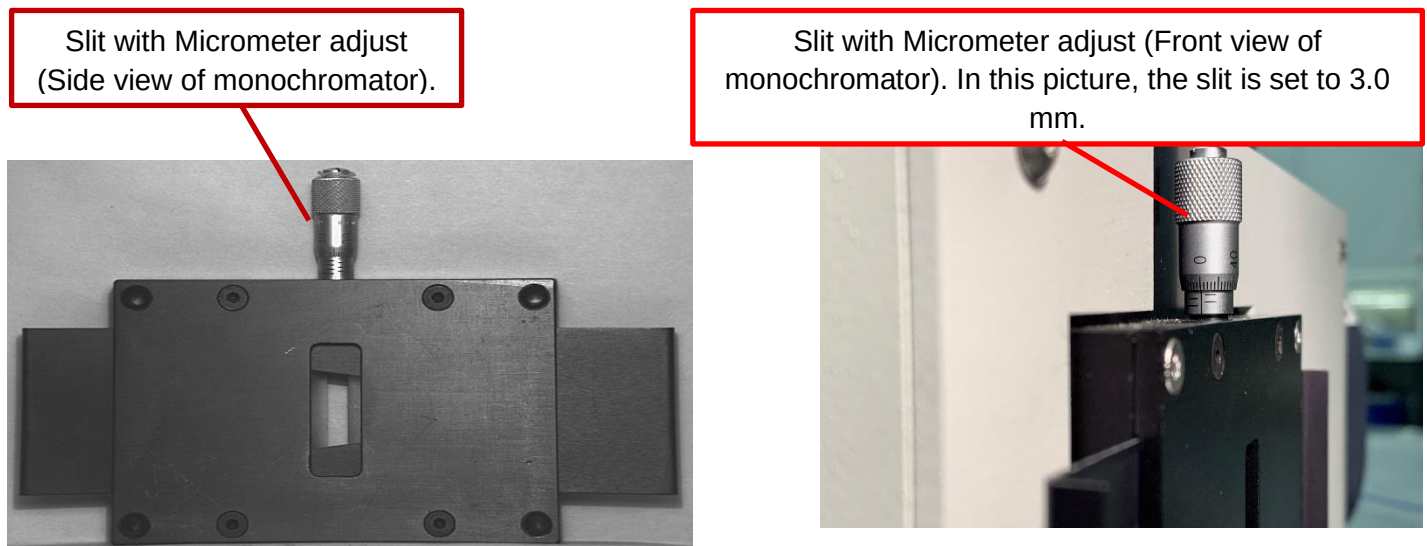


Figure 50. Slit with micrometer adjust.

For every 0.25 turns of the micrometer head from the closed position, the bandpass increases by 0.5 nm FWHM (for a 1200 line/mm pitch grating). For example, see the Table below:

Table 4: Micrometer slit width adjustment table.

Turns of Slit Micrometer Head	Bandpass (FWHM), nm With 1200l/mm Grating	Bandpass (FWHM), nm With 600l/mm Grating	Bandpass (FWHM), nm With 300l/mm Grating	Bandpass (FWHM), nm With 150l/mm Grating
0.25 (0.125 mm width)	0.5 (9055), 0.9 (9072)	1 (9055), 1.8 (9072)	2 (9055), 3.6 (9072)	4 (9055), 7.2 (9072)
0.50 (0.25 mm width)	1 (9055), 1.8 (9072)	2 (9055), 3.5 (9072)	4 (9055), 7 (9072)	8 (9055), 14 (9072)
1.00 (0.5 mm width)	2 (9055), 3.5 (9072)	4 (9055), 7.0 (9072)	8 (9055), 14 (9072)	16 (9055), 28 (9072)
2.00 (1 mm width)	4 (9055), 7.0 (9072)	8 (9055), 14 (9072)	16 (9055), 28 (9072)	32 (9055), 56 (9072)
4.00 (2 mm width)	8 (9055), 14 (9072)	16 (9055), 28 (9072)	32 (9055), 56 (9072)	64 (9055), 128 (9072)
12.00 (6 mm width)	24 (9055), 42 (9072)	48 (9055), 84 (9072)	96 (9055), 168 (9072)	192 (9055), 336 (9072)

Start the SciSpec software. The monochromator should be at position 0 nm (total specular reflection from the grating) after initializing and white light should be present at the output. Refer to the SciSpec Software Manual for details of controlling the monochromator and filter wheel.

Move the monochromator to a visible wavelength like 550nm (bright green). You should be able to clearly see the illumination leaving the tunable light source (except for a IR filament source as it does not produce enough visible radiation other than at 0nm).

13. Overall System Specifications

Table 5. Specifications and Certificates of Tunable Light Source systems

Model	TLS-72-X300	TLS-72-Q250	TLS-55-X300	TLS-55-Q250
See TR-TLS-01 for QC report of your system. If TLS is used as a part of PTS system, see TR-PTS-01				
General Specifications				
Lamp Type	Xe 300W	QTH 250W	Xe 300W	QTH 250W
Monochromator Type	9072S (1/8 meter)		9055 (1/4 meter)	
Functional Spectral Range	300 – 1800 nm			
Optical Resolution	0.5 nm @ 300-450 nm 0.7 nm @ 450-1800 nm		0.2 nm @ 300-450 nm 0.4 nm @ 450-1800 nm	
Wavelength Repeatability	0.1 nm		0.03 nm	
Wavelength Accuracy	0.2 nm			
Beam output	1 in [25 mm] diameter collimated			
Optical Height	8.5 in [215 mm], adjustable +/- 0.2 in [5 mm]			
Physical Specifications				
Typical Weight	30 kg		37 kg	
Dimensions of the TLS (L X W X H)	30.5 in [775 mm] x 16.4 in [416 mm] x 12.6 in [321 mm]		36.5 in [927 mm] x 21.5 in [542 mm] x 12.6 in [320 mm]	
Operating Environment	Indoor, well-ventilated environment. Not for household application.			
Operating Temperature Range	15 – 30 °C			
Operating Humidity Range	<80% RH, non-condensing			
Electrical Specifications				
Single Phase VAC Input				
Voltage (rms)	100 to 230 VAC (all components)			
Frequency	50/60 Hz (all components)			
Maximum Current (rms)	5.0 A (total all components)			
Fuses	6.25 A, 250 V (located inside the power supply)			
Lamp House Lamp Power				
Operating Voltage	15 – 20 VDC	0-24 VDC	15 – 20 VDC	0-24 VDC
Operating Current (maximum power)	15A	11A	15 – 20 VDC	11A
Monochromator Input Power				
Operating Voltage	24 VDC			
Maximum Current	0.5A			
Filter Wheel Input Power				
Operating Voltage	24 VDC			
Maximum Current	0.2A			
Certificates				
CE Certification	Yes, see Figure 51			

Specifications and Certificates of Tunable Light Source systems with DUAL light source.

Model	TLS-55-DUAL (Xenon and QTH)	TLS-55-DUAL (IR Filament and QTH)
See TR-TLS-01 for QC report of your system.		
General Specifications		
Lamp Type	Xe 300W, QTH 250W	QTH 250W, 37W IR Source
Monochromator Type	9055 (1/4 meter)	
Functional Spectral Range	250 – 2000 nm	400 – 5200 nm
Minimum Optical Resolution	0.2 nm, Grating 1	0.4 nm, Grating 1
	0.4 nm, Grating 2	0.8 nm, Grating 2
	0.8 nm, Grating 3	1.2 nm, Grating 3
Wavelength Repeatability	0.03 nm for Grating 1	0.06 nm for Grating 1
Wavelength Accuracy	0.2 nm for Grating 1	0.04 nm for Grating 1
Physical Specifications		
Typical Weight	140 lbs (64 kg)	
Dimensions of the TLS (L X W X H)	46.479 in [1181 mm] x 29.931 in [760 mm] x 13.190 in [335 mm]	
Operating Environment	Indoor, well-ventilated environment. Not for household application.	
Operating Temperature Range	15 – 30 °C	
Operating Humidity Range	<80% RH, non-condensing	
Electrical Specifications		
Single Phase VAC Input		
Voltage (rms)	100-240 VAC	
Frequency	50/60Hz	
Maximum Current (rms)	9A	
Fuses	6.25A, 250V (located inside the lamp power supplies)	
Lamp House Lamp Power		
Operating Voltage	15 – 20 VDC (LH-E-X300) 0-24 VDC (LH-S-Q250)	0-24 VDC
Operating Current (maximum power)	15 A (LH-E-X300) 11 A (LH-S-Q250)	11 A (LH-S-Q250) 1.5 A (LH-S-37IR)
Monochromator Input Power		
Operating Voltage	24 VDC	
Maximum Current	0.5A	
Filter Wheel Input Power		
Operating Voltage	24 VDC	
Maximum Current	0.2A	
Certificates		
CE Certification	Yes, see Figure 51	

14. EU Declaration of Conformity



EU Declaration of Conformity

In accordance with European Parliament and Council Decision No 768/2008/EC

European Union Directives: ☒ 2014/30/EU Electromagnetic Compatibility Directive (EMCD);
☒ 2014/35/EU Low Voltage Directive (LVD);
☐ 2012/19/EU Waste Electrical and Electronic Equipment (WEEE); ☒ R&D Only;
☒ 2011/65/EU Restriction of Hazardous Substances Directive (RoHS); ☐ R&D Only;

Harmonized Standards: ☒ EN 61236-1:2013 (EMC)(Note 1); ☐ Custom Permanent Installation (EMCD Clause 19.1);
☒ EN 61010-1:2010 (Safety);
☒ EN 62471:2008, IEC/TR 62471-2 (Photobiological Safety);

Manufacturer: Name: Sciencetech Inc.
Address: 1450 Global Drive, London, Ontario, Canada, N6N 1R3

Object of the declaration:
Equipment Type/Description: Tunable Light Source
Model Number(s) TLS-72-X300 [105-9001], TLS-55-300X [105-9002], TLS-72-250Q [105-9003],
TLS-55-250Q [105-9004], TLS-55-DUAL [Various]

Sciencetech Inc. declares its belief that the equipment listed above conforms to the selected European Union harmonized legislation. The listed equipment conforms to the selected directives and standards only if it is installed and operated in accordance with the manufacturer's specifications and instructions. This declaration of conformity is issued under the sole responsibility of the manufacturer.

Date: 28-Mar-2024

Scott Hafekost
Signature

Scott Hafekost

Full Name

President

Title

Note 1: Class A, Group 1

Figure 51. EU declaration of conformity for TLS systems.

15. Troubleshooting

Problem	Possible Cause	Recommended Action
Lamp housing's fan does not start	No power to the lamp housing	The 7W2 cable is not properly connected between the power supply and the lamp housing.
	Interlock fault	Interlock is not engaged. Make sure that lamp stage is all the way inserted into the lamp housing.
	Temperature Safety Switch is open	Contact Sciencetech. Due the overheating of the lamp house, the bimetallic temperature switch has been activated.
Lamp does not start	No power to the lamp house	The 7W2 cable is not properly connected between the power supply and the lamp housing.
	Defective fuse	Contact Sciencetech. The two fuses inside the power supply needs replacement with the same type and rating fuses.
	Old lamp	Replace the lamp with a new similar lamp. Contact Sciencetech for correct lamp specifications.
	Lamp not installed	No lamp installed. Avoid ignition. Disconnect supply before servicing or before lamp replacement.
	Incorrect lamp installation	Lamp is not installed correctly. Make sure the cathode and anode of the lamp (for xenon lamps) are connected to the correct adaptor pieces on the lamp stage and the wiring on the lamp stage are tight and firm.
	Ignition wire (for xenon lamps)	If the lamp is equipped with an ignition wire, make sure the ignition wire is intact and its place on the lamp.
	Insulation breakdown	Contact Sciencetech.
Lamp explosion	Old lamp	Replace the lamp. Lamp needs to be replaced with a similar lamp with correct type and rating. Contact Sciencetech for guidance.
	Overheating	Contact Sciencetech. The lamp became over heated and has been damaged. The lamp housing should be run in an indoor, well-ventilated environment with ambient temperature below 40°C and above 5°C.
	Contaminated lamp envelope	Replace the lamp. Do not handle/touch the lamp envelope. Handel the lamp with clean gloves and only by its anode and cathode ends. If the lamp envelope or contacts are contaminated, do not use the lamp.
	Defective lamp	Replace the lamp. Do not use a lamp that might have been exposed to mechanical or thermal shocks. Do not place the lamp on a hard surface like table top surface. Try to keep the lamp in its original box if it is not yet installed in the lamp housing.
Output instability	Old lamp	Replace the lamp. The lamp has been aged.
	Lamp house orientation	TLS systems is only supposed to be operated in orientations specified by Sciencetech.
Loss of power on the target plane	Dirty optics	Clean the optics without misaligning them.
	Issue with the filter wheel position	Contact Sciencetech
	Lamp misalignment	Contact Sciencetech
	Damaged optics	Contact Sciencetech.
	Aged lamp	Lamp has been aged. If the lamp is still in its safe average life time period, increase the power supply setting. Never go above 100% power supply setting.

16. Accessories

16.1 Off-axis Elliptical Reflector Installation (OAE)

The OAE output optic is removed for shipping. To attach the OAE, please follow these steps:

1. Loosen the both slip rings with a 3mm hex key indicated below.
2. Slide the optic fully in until the lens tube touches the bolt heads on the filter wheel.
3. Tighten both slip rings with a 3mm hex key indicated below.

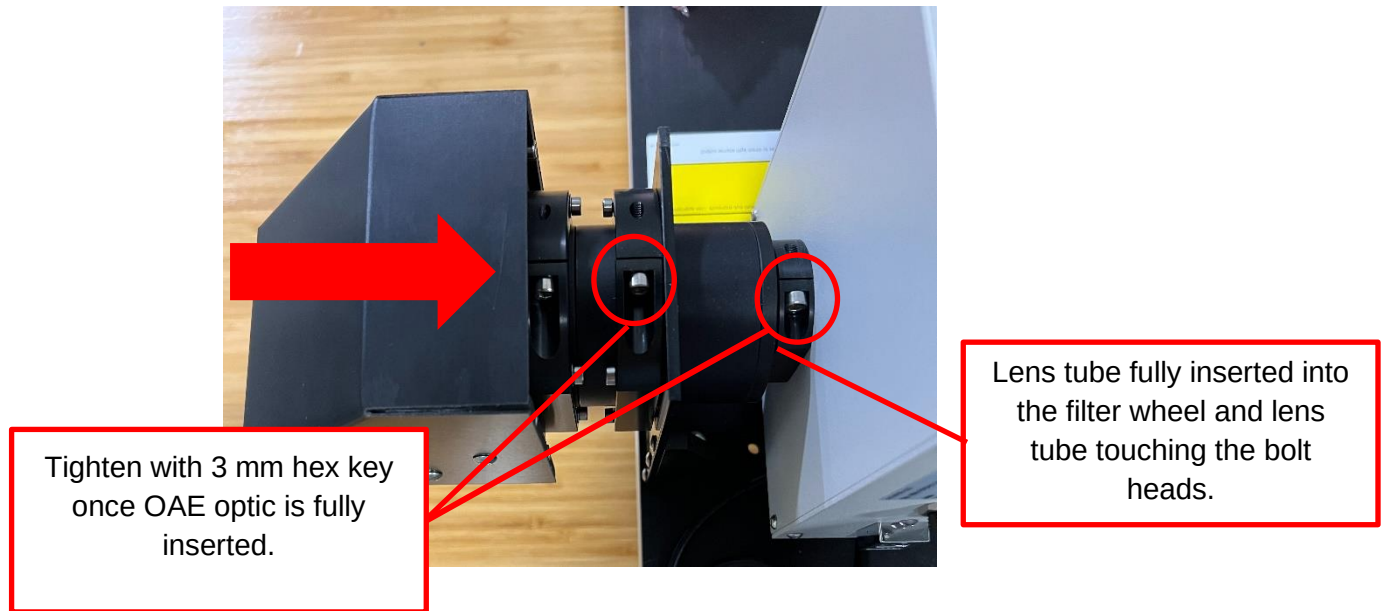


Figure 52: OAE installation.

16.2 Dual Input (DPIN-ST-MAN) and Output (DPOUT-ST-MAN) Ports for Sciencetech Standard Monochromators

This is upgrade to many of Sciencetech's "Standard" line of monochromators to allow use of a second input (if DPIN-ST-MAN purchased) or a second output ports (if DPOUT-ST-MAN purchased).

The heart of this functionality is a manual flip mirror that can be installed at input, output or both ports of the monochromators and provide the possibility to select between different input and output ports.

To turn the mirror from one position to the other (e.g. from side input (SIDE) to front input (FRONT) in Figure 53) follow these steps:



Figure 53. Push down the knurled knob the knob all the way down.

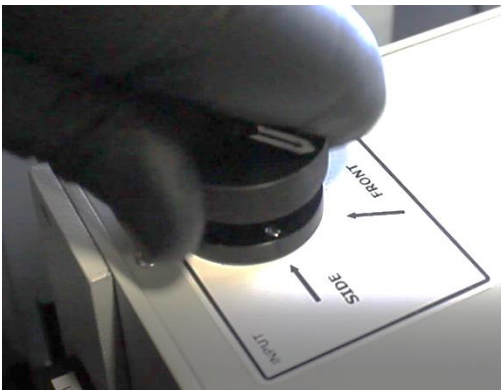


Figure 54. Turn the knob until the arrow on the knob align with the FRONT arrow on the label. You will feel a resistance at this location.



Figure 55. Slowly release the knob.

17. Appendix B. Power Optimization



Please note that all systems have been optimized and there is no need for power optimization of the system. If you were instructed to do power optimization or you like to do so because you have changed the lamp, follow these steps.

Make sure to follow all safety procedures described earlier in this user manual.

In order to optimize power, you need a detector and a data acquisition system. This is not provided by any TLS system by default. You may either use your own set or perhaps have purchased a set from Sciencetech to integrate with SciSpec software and TLS hardware. Please note that in some instances you may need an optical chopper as well.

By default, the beam coming out of a standard TLS system is somewhat collimated (~1" in diameter). As the active area of the majority of the detectors on the market is less than one inch, you may need a set of condenser optics to focus the light on the active area of the detector.

- 1- To start, attached the detector to the end of the condenser lens subassembly that you have attached to the output lens tube of the TLS system. Make sure to underfill the surface area of the detector.
- 2- Using the SciSpec, move to a wavelength such as 470nm. This way you can avoid saturating the detector. By default, the two slits on the monochromator are set to about 6mm opening. If you still saturate the detector, you can close bot slit on the monochromator by equal amount (for example to 1mm opening).

17.1 Power Optimization in TLS-72-X300 and TLS-55-X300 and 37W IR Source

-X300 and 37W IR Source TLS are using elliptical reflector to collect and focus the light from the lamp to the entrance of a monochromator. To optimize the power at the out of such TLS systems, follow these steps:

- 1- After performing setting in section 17, grab 2.5 mm hex driver and turn the three M3 screws at the rear of the lamp housing in one direction (CW or CCW), and by small equal amount (e.g ¼ of a turn). Using your detector and the data equitation system (DAQ), monitor if the power reading is increasing (Figure 56).

It is important to turn the lamp only in small equal steps to avoid touching the lamp by the reflector. This may also cause electrical hazard for you.

- 2- If the power increases in this direction of turn, continue truing the three screws by small equal amounts in the same direction, until you can not maximize the power reading on the DAQ system any more. At this point you might do very minute, one-time adjustment to individual screws to make sure there is maximum power is read on the DAQ.
- 3- If the power is not increasing in this direction (the direction you have choose in step 1), try performing the steps above in the other direction to (e.g. CCW) optimize direction.

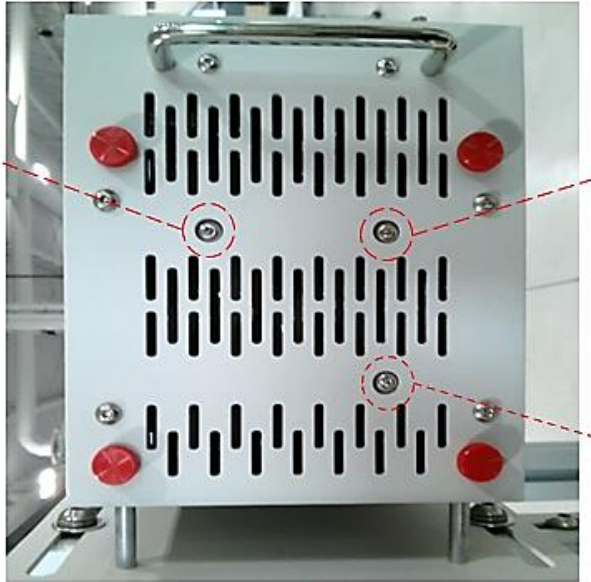


Figure 56. The rear of the TLS showing the three M3 lamp alignment screws.

17.2 Power Optimization in TLS-72-Q250 and TLS-55-Q250

TLS-72-Q250 and TLS-55-Q250 use a lamp housing with a spherical back reflector, a QTH lamp, and a set of spherical lenses to send a collimated beam into the monochromator. Follow these steps to optimize/maximize the output power. In these systems we assume the lamp is located at the correct location Infront of the lens. Thus, we only use adjustment on the spherical back reflector

- 1- After performing steps in section 17, grab a 2.5 hex drive and the rear spherical back reflector within the lamp house will be adjusted in very small. equal increments via three screws to maximize the signal. Turn the three M3 screws at the rear of the lamp housing in one direction (CW or CCW), and by small equal amount (e.g $\frac{1}{4}$ of a turn). Using your detector and the data equitation system (DAQ), monitor if the power reading is increasing.
- 2- If the power increase in this direction, continue truing by small equal amount in the same direction, until you can not maximize the power reading on the DAQ system any more. At this point you might do very minute adjustment to individual screws to make sure there is maximum power reading on the DAQ.
- 3- If the power is not increasing in this direction (the direction you have choose in step 1), try perfuming the steps above in the optimize direction



Figure 57. The rear of the TLS with QTH lamp showing the three M3 lamp alignment screws of the spherical back reflector.

18. Appendix C. Operating Adjustable Iris to Modify the Size of the Output Beam,

An adjustable iris might have been installed as a part of an add-on optical subassembly accessory at the output or within the optical train of the TLS system. Often the Iris has an adjustable ring with a locking screw to fix the iris at the desired aperture setting (Figure 58).

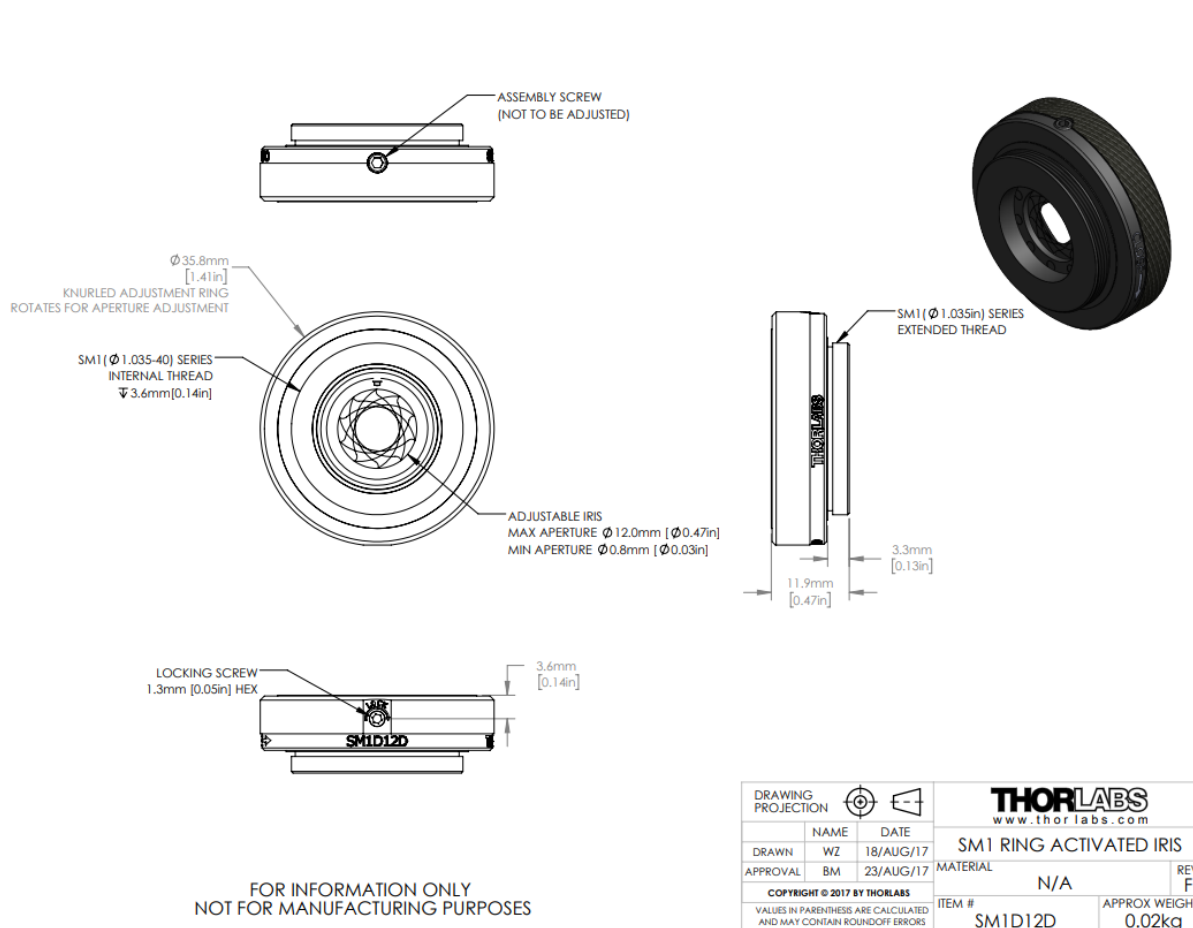


Figure 58. Technical drawing of the THORLABS SM1D1D iris indicating the location of the locking screw.

In some cases, the iris is graduated which gives the possibility to nominally set the aperture opening using the laser-engraved scale around its ring (Figure 59).

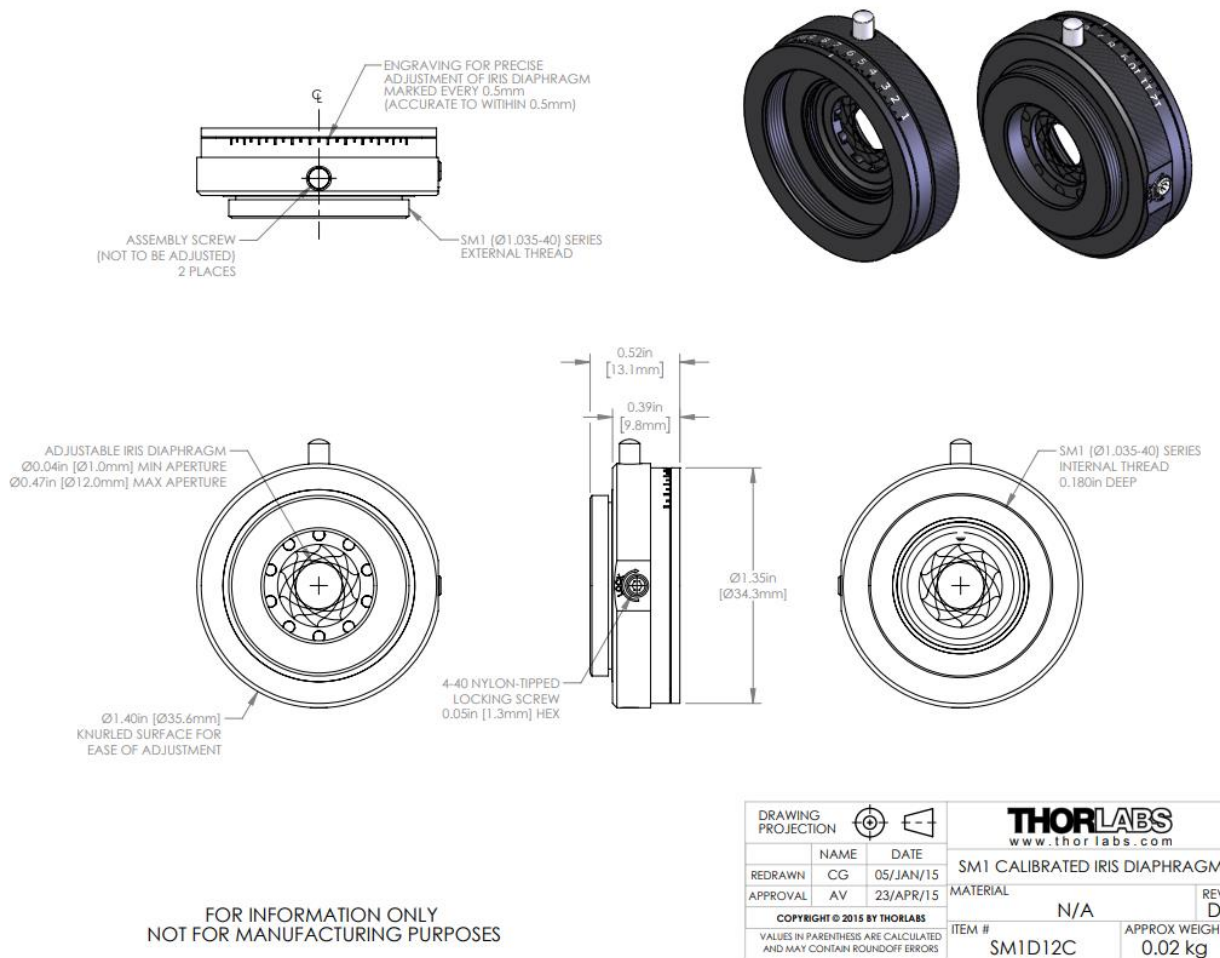


Figure 59. Technical drawing of the THORLABS SM1D1D iris indicating the location of the locking screw and the graduated ring with laser-engraved numbers.

To set the iris aperture follow these steps:

- 1- First use a 1.3x40 driver size to open the LOCKING SCREW of the iris.
- 2- Next, use the adjustable ring of the iris to change the size of the iris. If the iris had laser-engraved scale at the side of it, move the ring with laser engraved number on it to align the desired number in front of the straight line at the fixed ring of the iris. This will nominally set the aperture opening of the iris (Figure 60).

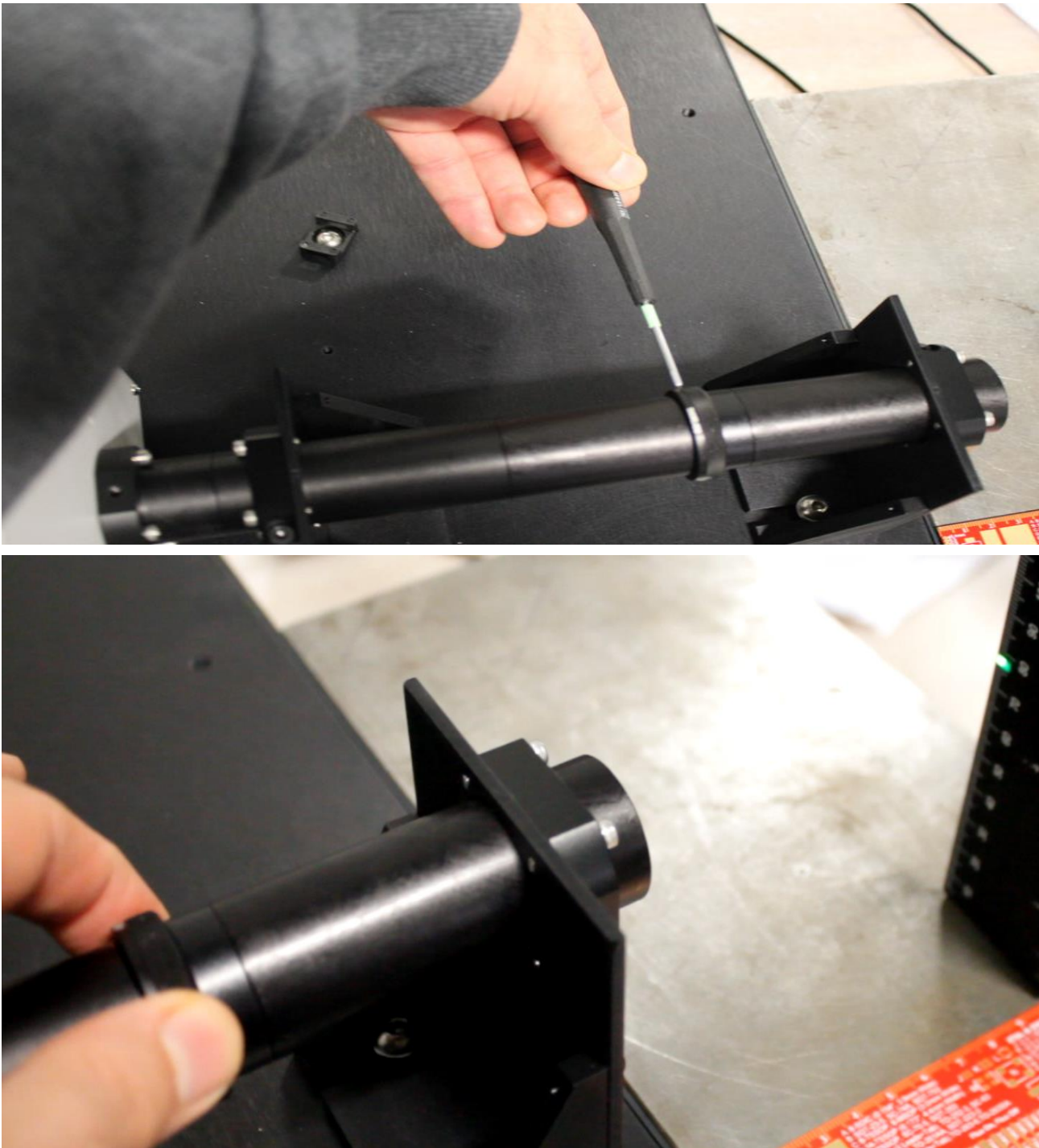


Figure 60. Unlocking and changing the aperture size of the adjustable iris to modify the size of the focused beam.

19. Important Notice

All electrical instruments may be hazardous if not handled in accordance with proper instructions and common precautions. Sciencetech Inc. will not be responsible for any damage caused by such units if instructions herein are not followed and repairs are not attended to or performed by company-trained or licensed personnel. All instruments should be operated with proper grounds on power line and should not be opened or handled as to electrical or electrically operated components without being switched off and disconnected from power receptacle.

Sciencetech Inc. reserves the right to make adjustments or improvements in its product without notice and without obligation to subsequent purchasers and without being required to make corresponding changes or improvements in products theretofore manufactured and sold.

We have done our very best in the manufacture and packing of this material. The transportation carrier is now responsible for delivering it to you in its original good condition, since all purchases are FOB London.

If the shipment is NOT delivered in good order and in accordance with quantity shown on Bill of Lading or Packing Slip, have the shortage or damage noted by the Carrier on both the delivery receipt and the freight bill, or by special form provided by United Parcel or the Post Office.

The Interstate Commerce Commission has ruled that Transportation Companies will not honor any losses or shortage claims unless exceptions are noted on the freight bill at the time of delivery. It is the buyer's responsibility to make a complete inspection immediately upon receipt of purchased goods.

If you accept shipment from the Transportation Carrier short of what is enumerated on the Bill of Lading – or in damaged condition – without having proper notation made by the Carrier, you do so at your own risk.

If bundles or crates are in apparent good order, but on opening contents are found to be damaged, call Carrier for adjuster to view same and have the Transportation Company/United Parcel/Post Office mark the freight bill or packing slip relative to such concealed damage. Make your claim at once for the Transportation Company/United Parcel/Post Office has a limited time for presentation of claims.

We are willing to assist you in every possible manner in collecting claims for loss or damage on this shipment, but this willingness on our part does not make us responsible for filing or collecting claims or replacing materials. Claims for Loss or Damage on shipment may not be deducted from our invoice, nor pay of the invoice withheld awaiting adjustment of such claims, as we cannot guarantee safe delivery.

Important: Do not return goods without written authority.

Contact factory for return material authorization.

Returned goods will not be accepted by us from the Transportation Company/United Parcel/Post Office unless written authorization has been issued by Sciencetech Inc.

Return of special or non-stock items cannot be authorized. Credit for goods returned - under authorization - will depend on the value to us based on our selling price, less a fair charge to cover the expense of shipping - re-handling - transportation - refinishing, etc., providing material is received in good condition - transportation charges prepaid - credit rendered to be used against future purchases.

All equipment manufactured by Sciencetech Inc. has been subjected to extensive performance and quality control testing. In order to constantly improve our product, we ask your assistance. Upon installation of our equipment, please fill out the attached card and return to us.

By completing the card and returning it to Sciencetech, you will register your instrument in warranty and enable us to provide you with the best possible service.

20. Warranty and Assistance

Sciencetech's warranty and return policy is subject to change.

If you have any questions or issues related to the operation of your Tuneable Light Source and its accessories, please contact:

Sciencetech Inc.
1450 Global Dr.
London, ON N6N 1R3, Canada
Telephone: 519-644-0135
Fax: 519-644-0136
Email: sales@sciencetech-inc.com

For our up-to-date contact and warranty information please see:

<http://www.sciencetech-inc.com/warranty-returns>

If your shipment is received in damaged or poor condition, please see Sciencetech's detailed policies on how to proceed on the Sciencetech website